Describe a way to choose a random sample of 50 students from a school list.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

2401

QUESTION ID

S65-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Number the students and use a random number generator to pick 50 different numbers

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is surveying students in the cafeteria line random?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

2402

QUESTION ID

S65-B3-Q2

TYPE

Short answer

CORRECT ANSWER

No: not every student has an equal chance

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q2/5

True or false?

A random sample tends to represent the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

2403

QUESTION ID

S65-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE No systematic bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sample is random?

A. friends of the teacher

B. students who volunteer

C. names drawn from a hat with every student's name

D. the first 50 arrivals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

2404

QUESTION ID

S65-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. names drawn from a hat with every student's name

GRADING NOTE Chance selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does a survey need a random sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

2405

QUESTION ID

S65-B3-Q5

TYPE

Short answer

CORRECT ANSWER

To generalize its results to the population

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 65 • BACKWARD 3 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 1

Identify confounding and design flaws • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Students who choose to take an SAT prep course score higher. Give a confounding variable and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

2407

QUESTION ID
S65-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Motivation (or income): students who choose the course may differ in ways that also raise scores.

GRADING NOTE Any valid confounder with explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Students who choose to take an SAT prep course score higher than students who do not. Redesign the study so it can show whether the course causes higher scores.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

2408

QUESTION ID
S65-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Randomly assign volunteers to take the course or not, then compare scores.

GRADING NOTE Any valid experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q2/5

True or false?

An experiment without a control group can still show the treatment caused the result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

2409

QUESTION ID

S65-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Comparison is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q3/5

In an experiment, why use a placebo?

A. to make the study observational

B. to control for the effect of believing one is treated

C. to increase the sample size

D. to make the treatment stronger

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

2410

QUESTION ID

S65-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. to control for the effect of believing one is treated

GRADING NOTE Placebo effect.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why might an observational study be chosen instead of an experiment for smoking?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

2411

QUESTION ID
S65-F1-Q5

TYPE
Short answer

CORRECT ANSWER
It would be unethical to assign people to smoke.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 65 • FORWARD 1 • Q5/5

SET 65 • HSS-IC.B.3

FORWARD 2

Analyze the results of an experiment • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

100 volunteers were randomly assigned to a vitamin or a placebo. The vitamin group had 1.2 fewer colds on average. A re-randomization found 40 of 200 differences at least that large. What do you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

2413

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Not significant: 20% of re-randomizations give a difference this large by chance

GRADING NOTE Form required: conclusion and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q1/5

In a randomized vitamin-versus-placebo experiment, the vitamin group had 1.2 fewer colds on average, and only 2 of 200 re-randomized differences were that large. What do you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

2414

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Significant: the vitamin likely reduces colds

GRADING NOTE Form required: conclusion and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q2/5

True or false?

Random assignment makes it possible to attribute a significant difference to the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

2415

QUESTION ID
S65-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Groups differ only by chance otherwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q3/5

The re-randomization model assumes

A. the groups are different sizes

B. the treatment has no effect

C. the treatment works

D. the data are normal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

2416

QUESTION ID

S65-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the treatment has no effect

GRADING NOTE Null model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q4/5

Explain why an observational study cannot be analyzed in the same causal way.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

2417

QUESTION ID

S65-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Without random assignment, a difference could come from confounding variables, not the treatment.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 65 • FORWARD 2 • Q5/5

SET 66 • HSS-IC.B.3 • STATISTICS, INFERENCE, AND DECISIONS

Explain random sampling versus random assignment

MAIN	Algebra II main question	**HSS-IC.B.3**
BACKWARD 1	Study types and their purposes	**HSS-IC.B.3**
BACKWARD 2	Choose a random sample	**7.SP.A.1**
BACKWARD 3	Correlation and causation	**HSS-ID.C.9**
FORWARD 1	Improve an experimental design	**HSS-IC.B.3**
FORWARD 2	Use re-randomization to judge a treatment difference	**HSS-IC.B.5**

SET 66 • HSS-IC.B.3

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A random sample of 500 adults is surveyed about exercise and sleep; more exercise is associated with better sleep. What can and cannot be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

2420

QUESTION ID

S66-M-Q1

TYPE

Short answer

CORRECT ANSWER

The association can be generalized to all adults (random sample), but not that exercise causes better sleep (no random assignment)

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers are randomly assigned to two diets; diet A loses more weight. What can and cannot be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

2421

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

Diet A caused more weight loss for these volunteers (random assignment), but it cannot be generalized to all people (not a random sample)

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q2/5

A random sample of 200 students is randomly assigned to two study methods. What can be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

2422

QUESTION ID

S66-M-Q3

TYPE

Short answer

CORRECT ANSWER

Both: cause and effect, generalized to the student population

GRADING NOTE Form required: both conclusions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q3/5

A teacher compares test scores of students who chose to use a review app with those who did not, in her own class. What can be concluded?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

2423

QUESTION ID

S66-M-Q4

TYPE

Short answer

CORRECT ANSWER

Neither causation nor generalization: no random assignment and no random sample; only an association in this class

GRADING NOTE Form required: the conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q4/5

Describe how to randomly assign 40 subjects to two groups of 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

2424

QUESTION ID

S66-M-Q5

TYPE

Short answer

CORRECT ANSWER

Number the subjects 1-40; use a random number generator to choose 20 different numbers for group 1; the rest form group 2

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-M-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • MAIN • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 1

Study types and their purposes • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Which type of study uses random assignment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

2426

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

An experiment

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which type of study is designed to estimate a population value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

2427

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

A sample survey

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An observational study assigns treatments.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

2428

QUESTION ID

S66-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It observes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Random assignment is used to

A. avoid measuring

B. make the sample larger

C. make the groups comparable

D. choose who is in the population

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

2429

QUESTION ID

S66-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. make the groups comparable

GRADING NOTE Comparability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What does random sampling help you do?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

2430

QUESTION ID

S66-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Generalize to the population

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • BACKWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 2

Choose a random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Describe how to choose 20 names at random from a list of 500.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

2432

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Number the names and pick 20 different random numbers from 1 to 500

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is a sample of volunteers random?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

2433

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A sample of volunteers may differ from the population.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

2434

QUESTION ID

S66-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Self-selection.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q3/5

Which method gives a random sample?

A. choosing the first 20

B. drawing numbered slips from a well-mixed hat

C. asking for volunteers

D. asking friends

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

2435

QUESTION ID

S66-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. drawing numbered slips from a well-mixed hat

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why must slips in a hat be well mixed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

2436

QUESTION ID

S66-B2-Q5

TYPE

Short answer

CORRECT ANSWER

So every slip has the same chance of being drawn

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • HSS-IC.B.3

BACKWARD 3

Correlation and causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

ANSWER KEY Reduced copy of the question page, with its answer

An association between two variables in observational data: can it prove causation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

2438

QUESTION ID

S66-B3-Q1

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give a possible lurking variable for "exercise and sleep".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

2439

QUESTION ID

S66-B3-Q2

TYPE

Short answer

CORRECT ANSWER

For example age or stress

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q2/5

True or false?

Random assignment controls lurking variables on average.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

2440

QUESTION ID
S66-B3-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Balances groups.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Causation can be concluded from

A. an observational study

B. a correlation alone

C. a randomized experiment

D. a survey

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q4/5

2441

QUESTION ID

S66-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q4/5

Why does a correlation not show which variable causes the other?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

2442

QUESTION ID

S66-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Either could affect the other, or a third variable could affect both

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S66-B3-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 66 • BACKWARD 3 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 1

Improve an experimental design • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers try a new diet and lose weight. Explain why a control group (for example a second, randomly assigned diet) is needed to judge the diet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

2444

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Without a comparison group we cannot tell whether weight loss was due to the diet or to other factors.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug study blocks subjects by age before randomly assigning treatments. Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

2445

QUESTION ID
S66-F1-Q2

TYPE
Short answer

CORRECT ANSWER
To make sure age differences do not hide or create a treatment effect; groups are compared within similar ages.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q2/5

True or false?

In a double-blind experiment, neither subjects nor evaluators know who gets the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

2446

QUESTION ID

S66-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers were randomly assigned to two diets, and diet A led to more weight loss. Which change would let the result be generalized to all adults?

A. measuring more often

B. using a placebo

C. adding more volunteers

D. randomly selecting the subjects from all adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

2447

QUESTION ID
S66-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. randomly selecting the subjects from all adults

GRADING NOTE Random sampling.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q4/5

Why are many subjects better than few in an experiment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

2448

QUESTION ID
S66-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Chance differences between the groups are smaller, so real effects are easier to detect.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 66 • FORWARD 1 • Q5/5

SET 66 • HSS-IC.B.3

FORWARD 2

Use re-randomization to judge a treatment difference • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

60 volunteers were randomly assigned to two diets; diet A lost 2.4 kg more on average. In 200 re-randomizations, 5 differences were at least 2.4 kg. Is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

2450

QUESTION ID

S66-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: 2.5% of re-randomizations, which is unusual

GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

60 volunteers were randomly assigned to two diets, and diet A lost significantly more weight. State the conclusion in context, including to whom it applies.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

2451

QUESTION ID

S66-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Diet A caused more weight loss for these volunteers; it may not apply to all adults.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q2/5

True or false?

Re-randomization shuffles the group labels as if the treatment had no effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

2452

QUESTION ID
S66-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Null model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 70 of 200 re-randomized differences were at least 2.4 kg, conclude

A. the difference could be due to chance

B. the study was observational

C. diet B works

D. diet A works

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

2453

QUESTION ID
S66-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. the difference could be due to chance

GRADING NOTE 35% is common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why re-randomization is valid only because the original assignment was random.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

2454

QUESTION ID

S66-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Shuffling copies the original random assignment process; without it, the shuffle does not model how the groups were formed.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 66 • FORWARD 2 • Q5/5

SET 67 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population proportion with a margin of error from a simulation

MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Statistics, parameters, and sampling variability	**HSS-IC.A.1**
BACKWARD 2	Proportions and percents	**6.RP.A.3.c**
BACKWARD 3	Standard deviation as typical distance from the mean	**HSS-ID.A.2**
FORWARD 1	Plan a sample size for a desired margin of error	**HSS-IC.B.4**
FORWARD 2	Evaluate reports that use margins of error	**HSS-IC.B.6**

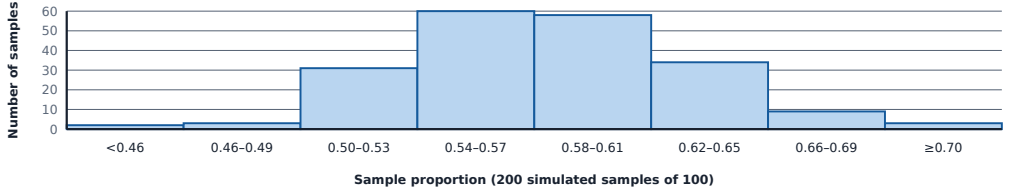
SET 67 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58 want a later start time. The histogram shows 200 simulated samples of 100 from a population with p = 0.58; their standard deviation is 0.049. Give a margin of error and an interval estimate.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

2457

QUESTION ID

S67-M-Q1

TYPE

Short answer

CORRECT ANSWER

Margin about 2(0.049) ≈ 0.10; interval about 0.48 to 0.68

GRADING NOTE Form required: margin and interval (0.098 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 300 voters, 120 support a measure. Simulated sample proportions (n = 300, p = 0.40) have standard deviation 0.0275. Give the estimate, the margin of error, and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

2458

QUESTION ID
S67-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.40 ± 0.055; about 0.345 to 0.455

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q2/5

In a random sample of 50 shoppers, 35 use a store app. Simulated proportions (n = 50, p = 0.70) have standard deviation 0.064. Give the margin of error and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

2459

QUESTION ID

S67-M-Q3

TYPE

Short answer

CORRECT ANSWER

About 0.13; about 0.57 to 0.83

GRADING NOTE Form required: both (0.128 accepted).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q3/5

Of the three surveys (n = 50, 100, 300), which has the smallest margin of error, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

2460

QUESTION ID

S67-M-Q4

TYPE

Short answer

CORRECT ANSWER

n = 300: larger samples produce less variable sample proportions

GRADING NOTE Form required: answer and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "52% of voters support the plan, with a margin of error of 3 percentage points." Interpret this statement.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

2461

QUESTION ID
S67-M-Q5

TYPE
Short answer

CORRECT ANSWER
The true percent of all voters who support the plan is plausibly between 49% and 55%

GRADING NOTE Any equivalent interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • MAIN • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 1

Statistics, parameters, and sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58 want a later start time. What is the statistic?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

2463

QUESTION ID
S67-B1-Q1

TYPE
Short answer

CORRECT ANSWER
0.58 (58%)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 100 students, 58% want a later start time. What parameter does 58% estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

2464

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The proportion of all students who want a later start

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q2/5

True or false?

Different random samples of 100 would give different sample proportions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

2465

QUESTION ID

S67-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q3/5

A histogram of 200 simulated sample proportions (samples of 100) shows

A. the parameter exactly

B. how sample proportions vary from sample to sample

C. the population

D. one sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

2466

QUESTION ID

S67-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. how sample proportions vary from sample to sample

GRADING NOTE Sampling distribution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q4/5

Why does a random sample matter for the margin of error to be meaningful?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

2467

QUESTION ID

S67-B1-Q5

TYPE

Short answer

CORRECT ANSWER

The margin describes chance variation in random samples; a biased sample can be off by more.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 67 • BACKWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 2

Proportions and percents • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Write 120 out of 300 as a proportion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

2469

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.40

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 0.58 ± 0.10 as an interval?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

2470

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.48 to 0.68

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

35 out of 50 is 35%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

2471

QUESTION ID

S67-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 35/50 = 0.70 = 70%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

52% ± 3% gives

A. 52% to 55%

B. 49% to 55%

C. 50% to 54%

D. 49% to 52%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

2472

QUESTION ID

S67-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 49% to 55%

GRADING NOTE Subtract and add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.055 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

2473

QUESTION ID

S67-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 67 • BACKWARD 2 • Q5/5

SET 67 • HSS-IC.B.4

BACKWARD 3

Standard deviation as typical distance from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

About what percent of mound-shaped data lie within 2 standard deviations of the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

2475

QUESTION ID
S67-B3-Q1

TYPE
Short answer

CORRECT ANSWER
About 95%

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

If the SD is 0.049, what is 2 SDs?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

2476

QUESTION ID

S67-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0.098

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B3-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q2/5

True or false?

A smaller standard deviation means more variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

2477

QUESTION ID

S67-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A smaller SD means the values are closer to the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which set of simulated proportions varies least?

A. SD 0.1

B. SD 0.0275

C. SD 0.049

D. SD 0.064

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

2478

QUESTION ID

S67-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. SD 0.0275

GRADING NOTE Smallest SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-B3-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 0.064.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

2479

QUESTION ID

S67-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.128

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 67 • BACKWARD 3 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 1

Plan a sample size for a desired margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A survey of 100 students has a margin of error of about 0.10. Quadrupling the sample size roughly halves the margin of error. Estimate the margin with n = 400.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

2481

QUESTION ID

S67-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 0.05

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

About what sample size gives a margin of error of 0.05 when $p \approx 0.5$? (Use margin $\approx 2\sqrt{(p(1-p)/n)}$.)

$n \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

2482

QUESTION ID
S67-F1-Q2

TYPE
Short answer

CORRECT ANSWER
400

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q2/5

True or false?

Doubling the sample size halves the margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q3/5

2483

QUESTION ID

S67-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It divides it by √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q3/5

Which change most reduces the margin of error?

A. multiplying n by 9

B. doubling n

C. adding 10 people

D. repeating the same sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

2484

QUESTION ID

S67-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. multiplying n by 9

GRADING NOTE Margin divides by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q4/5

Two polls of the same population have margins 0.05 and 0.025. How do their sample sizes compare?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

2485

QUESTION ID

S67-F1-Q5

TYPE

Short answer

CORRECT ANSWER

The second is about 4 times as large

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S67-F1-Q5**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 67 • FORWARD 1 • Q5/5

SET 67 • HSS-IC.B.4

FORWARD 2

Evaluate reports that use margins of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A poll shows candidate A at 51% and B at 49% with margin ±4%. A headline says "A leads." Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

2487

QUESTION ID

S67-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Not justified: the intervals overlap widely, so either could lead

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report gives 58% ± 10% from 100 students and claims "a majority of students want a later start." Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

2488

GRADING NOTE Form required: evaluation and reason.

QUESTION ID

S67-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Not clearly justified: the interval 48% to 68% includes values below 50%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q2/5

True or false?

A margin of error accounts for bias from a voluntary-response sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

2489

QUESTION ID
S67-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Only chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q3/5

Which report is best supported?

A. 45% ± 2% so a majority oppose

B. 70% ± 3% support, from a random sample

C. 51% ± 5% support, so most support

D. 60% support from an online poll

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

2490

QUESTION ID

S67-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 70% ± 3% support, from a random sample

GRADING NOTE The interval stays above 50% and the sample is random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What should a careful report include besides the percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

2491

QUESTION ID
S67-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The sample size, how the sample was chosen, and the margin of error

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S67-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 67 • FORWARD 2 • Q5/5

SET 68 • HSS-IC.B.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate a population mean with a margin of error from a simulation

MAIN	Algebra II main question	**HSS-IC.B.4**
BACKWARD 1	Mean of a data set	**6.SP.B.5.c**
BACKWARD 2	What resampling simulates	**HSS-IC.A.1**
BACKWARD 3	Two standard deviations and the 95% range	**HSS-ID.A.2**
FORWARD 1	Use intervals to judge claims about a mean	**HSS-IC.B.4**
FORWARD 2	Compare two group means with re-randomization	**HSS-IC.B.5**

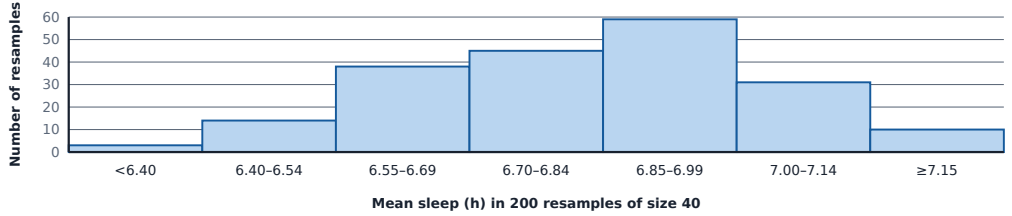
SET 68 • HSS-IC.B.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 40 students slept a mean of 6.8 hours. The histogram shows the means of 200 resamples of the data; their standard deviation is 0.20 hours. Give a margin of error and an interval estimate.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

2494

QUESTION ID

S68-M-Q1

TYPE

Short answer

CORRECT ANSWER

Margin about 0.4 h; interval about 6.4 to 7.2 hours

GRADING NOTE Form required: margin and interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 25 apples has mean weight 152 g. Simulated sample means have standard deviation 3.6 g. Give the margin of error and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

2495

QUESTION ID

S68-M-Q2

TYPE

Short answer

CORRECT ANSWER

About 7.2 g; about 144.8 g to 159.2 g

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 60 commuters has mean commute 24.5 min. Simulated means have standard deviation 1.4 min. Give the margin of error and the interval.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

2496

QUESTION ID

S68-M-Q3

TYPE

Short answer

CORRECT ANSWER

About 2.8 min; about 21.7 to 27.3 minutes

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q3/5

A random sample of 40 students gives the interval 6.4 to 7.2 hours for the mean hours of sleep. Interpret the interval correctly, and explain why it does not mean that 95% of students sleep between 6.4 and 7.2 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

2497

QUESTION ID

S68-M-Q4

TYPE

Short answer

CORRECT ANSWER

The mean sleep of all students is plausibly between 6.4 and 7.2 hours (not: 95% of students sleep that long)

GRADING NOTE Form required: a correct interpretation contrasted with the individual-value misreading.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 25 apples gives a margin of error of 7.2 g for the mean weight. About what margin of error do you expect from a random sample of 100 apples?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

2498

QUESTION ID
S68-M-Q5

TYPE
Short answer

CORRECT ANSWER
About 3.6 g (half)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • MAIN • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 1

Mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 6, 7, 7.5, 6.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

2500

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6.75

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What does a sample mean estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

2501

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The population mean

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The mean is not affected by extreme values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

2502

QUESTION ID

S68-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE One extreme value can move the mean a lot; the median is resistant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The mean of 150, 152, 154 is

A. 154

B. 150

C. 152

D. 456

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

2503

QUESTION ID

S68-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 152

GRADING NOTE Sum/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

If each of 40 values increases by 0.5, how does the mean change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

2504

QUESTION ID

S68-B1-Q5

TYPE

Short answer

CORRECT ANSWER

It increases by 0.5

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 68 • BACKWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 2

What resampling simulates • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Why can resampling the data show how sample means vary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

2506

QUESTION ID

S68-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Resamples act like new random samples from a population that looks like the sample

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a histogram of 200 resample means, what does one bar's height mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

2507

QUESTION ID

S68-B2-Q2

TYPE

Short answer

CORRECT ANSWER

How many of the 200 resample means fell in that interval

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Sample means vary less than individual values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

2508

QUESTION ID

S68-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Averaging reduces variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 40 students slept a mean of 6.8 hours. The parameter being estimated is

A. 6.8

B. the mean sleep of all students

C. the 40 students' mean

D. the resample SD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

2509

QUESTION ID

S68-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the mean sleep of all students

GRADING NOTE Population mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 6.8 hours a parameter or a statistic?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

2510

QUESTION ID

S68-B2-Q5

TYPE

Short answer

CORRECT ANSWER

A statistic

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • HSS-IC.B.4

BACKWARD 3

Two standard deviations and the 95% range • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Compute 6.8 − 2(0.20) and 6.8 + 2(0.20).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

2512

QUESTION ID

S68-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6.4 and 7.2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 2 × 1.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

2513

QUESTION ID

S68-B3-Q2

TYPE

Short answer

CORRECT ANSWER

2.8

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q2/5

True or false?

About 95% of a mound-shaped distribution is within 2 SDs of its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

2514

QUESTION ID

S68-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

152 ± 7.2 gives

A. 144.8 to 159.2

B. 145 to 159

C. 152 to 159.2

D. 144.8 to 152

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

2515

QUESTION ID

S68-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 144.8 to 159.2

GRADING NOTE Subtract and add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S68-B3-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 7.5 hours inside the interval 6.4 to 7.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

2516

QUESTION ID

S68-B3-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 68 • BACKWARD 3 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 1

Use intervals to judge claims about a mean • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A random sample gives the interval 6.4 to 7.2 hours for students' mean sleep. A health guide says students average 8 hours. Does the interval support this? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

2518

QUESTION ID

S68-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No: 8 is well outside the plausible values

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q1/5

2518

A random sample gives the interval 144.8 g to 159.2 g for the mean apple weight. A grower claims apples average 155 g. Is the claim consistent with the interval?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

2519

QUESTION ID

S68-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes: 155 is inside the interval

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q2/5

True or false?

A value inside the interval is proved to be the true mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

2520

QUESTION ID

S68-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is only plausible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 60 commuters gives an interval for the mean commute time. Which change would most likely narrow the interval?

A. sampling 240 commuters instead of 60

B. using the median

C. rounding the data

D. sampling 30 commuters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

2521

QUESTION ID
S68-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. sampling 240 commuters instead of 60

GRADING NOTE Larger n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of students at one school gives the interval 6.4 to 7.2 hours for mean sleep. Explain why the interval should not be applied to students at a different school.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

2522

QUESTION ID
S68-F1-Q5

TYPE
Short answer

CORRECT ANSWER
The sample came only from this school's students, so it estimates their mean only.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 68 • FORWARD 1 • Q5/5

SET 68 • HSS-IC.B.4

FORWARD 2

Compare two group means with re-randomization • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In an experiment, 20 students using a study app scored 6 points more on average than 20 without it. In 200 re-randomizations, 9 differences were at least 6. Is the difference significant at the 5% level?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

2524

GRADING NOTE Form required: decision and proportion.

QUESTION ID

S68-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes:

9/200 = 4.5% < 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What model do the re-randomizations represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

2525

QUESTION ID
S68-F2-Q2

TYPE
Short answer

CORRECT ANSWER
The app has no effect on scores

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q2/5

True or false?

An interval for one mean and a re-randomization test for two means both use simulated variability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

2526

QUESTION ID
S68-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Same idea.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 30 of 200 differences were at least 6, the conclusion is

A. the difference could be chance

B. the app works

C. the study failed

D. the app hurts scores

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

2527

QUESTION ID

S68-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the difference could be chance

GRADING NOTE 15%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the margin of error for one mean is related to judging a difference of two means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

2528

QUESTION ID
S68-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Both ask how much a statistic varies by chance; a difference is meaningful only if it is large compared with that variability.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S68-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 68 • FORWARD 2 • Q5/5

SET 69 • HSS-IC.B.5 • STATISTICS, INFERENCE, AND DECISIONS

Compare two treatments with a re-randomization simulation

MAIN	Algebra II main question	**HSS-IC.B.5**
BACKWARD 1	Compare the means of two groups	**7.SP.B.4**
BACKWARD 2	Random assignment in experiments	**HSS-IC.B.3**
BACKWARD 3	Probability as relative frequency in simulations	**7.SP.C.6**
FORWARD 1	Two-sided comparisons and design choices	**HSS-IC.B.5**
FORWARD 2	Evaluate reports of experiments	**HSS-IC.B.6**

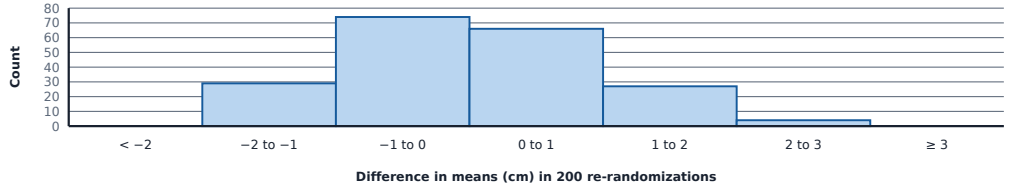
SET 69 • HSS-IC.B.5

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

10 plants got fertilizer and 10 did not (randomly assigned). Fertilized plants grew 2.9 cm more on average. The histogram shows 200 re-randomized differences; none was 2.9 or more. Is the difference significant? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

2531

QUESTION ID

S69-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes: a difference of 2.9 cm almost never happens by chance alone (0 of 200), so the fertilizer appears to increase growth

GRADING NOTE Form required: decision and evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a caffeine experiment, the caffeine group's reaction time was 0.04 s faster on average. In 200 re-randomizations, 62 differences were at least 0.04 s. Is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

2532

QUESTION ID
S69-M-Q2

TYPE
Short answer

CORRECT ANSWER
No: 31% of re-randomizations give a difference that large by chance

GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Students randomly assigned to a study app scored 6 points higher on average. In 200 re-randomizations, 9 differences were at least 6. Is the difference significant at the 5% level?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

2533

QUESTION ID

S69-M-Q3

TYPE

Short answer

CORRECT ANSWER

Yes: 9/200 = 4.5%, below 5%

GRADING NOTE Form required: decision and proportion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q3**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q3/5

Explain what a re-randomization simulation assumes and how it is carried out.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

2534

QUESTION ID

S69-M-Q4

TYPE

Short answer

CORRECT ANSWER

It assumes the treatment has no effect; all responses are pooled, randomly re-split into two groups of the original sizes, and the difference in means is recorded, many times.

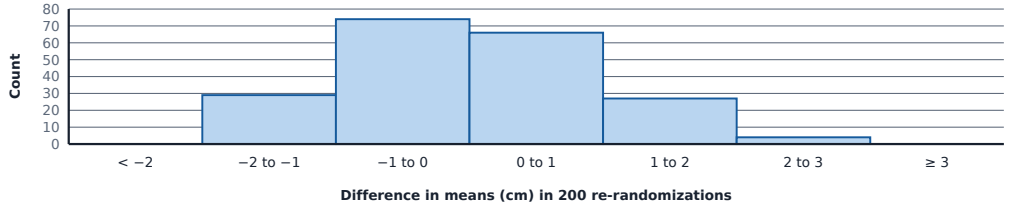
GRADING NOTE Any equivalent explanation with assumption and procedure.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the histogram of 200 re-randomized differences, about how many were at least 2 cm? Would an observed difference of 2 cm be significant at the 5% level?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

2535

QUESTION ID

S69-M-Q5

TYPE

Short answer

CORRECT ANSWER

4 (the 2-to-3 bar); 4/200 = 2%, so yes

GRADING NOTE Form required: count, proportion, decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • MAIN • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 1

Compare the means of two groups • Grade 7 • Statistics & Probability • 7.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Group A has mean 25.4 cm and group B has mean 22.5 cm. Find the difference A − B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

2537

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2.9 cm

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Group A: 4, 6, 8. Group B: 3, 5, 4. Find the difference in means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

2538

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q2/5

True or false?

A difference in sample means could happen even if the treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

2539

QUESTION ID

S69-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Mean A = 0.52 s and mean B = 0.48 s. A − B is

A. −0.04 s

B. 0.4 s

C. 1.00 s

D. 0.04 s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

2540

QUESTION ID

S69-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.04 s

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q4/5

Why compare the difference with the variability in the groups?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

2541

QUESTION ID

S69-B1-Q5

TYPE

Short answer

CORRECT ANSWER

A difference is meaningful only if it is large relative to the natural variation

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 2

Random assignment in experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

In a fertilizer experiment, 20 plants were randomly assigned to fertilizer or no fertilizer. Why were they randomly assigned?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

2543

QUESTION ID
S69-B2-Q1

TYPE
Short answer

CORRECT ANSWER
To make the groups comparable so differences can be attributed to the fertilizer

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Describe how to randomly assign 20 plants into two groups of 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

2544

QUESTION ID

S69-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Number the plants 1-20 and use random numbers to choose 10 for fertilizer

GRADING NOTE Any valid method.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q2/5

True or false?

Without random assignment, a re-randomization test does not justify a cause-and-effect conclusion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

2545

QUESTION ID

S69-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Design matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Random assignment is a feature of

A. censuses

B. experiments

C. observational studies

D. surveys

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

2546

QUESTION ID

S69-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. experiments

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q4/5

What kind of conclusion does a significant result in a randomized experiment allow?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

2547

QUESTION ID

S69-B2-Q5

TYPE

Short answer

CORRECT ANSWER

That the treatment caused the difference (for these subjects)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • HSS-IC.B.5

BACKWARD 3

Probability as relative frequency in simulations • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

62 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

2549

QUESTION ID

S69-B3-Q1

TYPE

Short answer

CORRECT ANSWER

31%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

9 of 200 is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

2550

QUESTION ID

S69-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q2/5

True or false?

An event that happens in 40% of simulated runs is unusual.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

2551

QUESTION ID

S69-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 40% is common; unusual usually means about 5% or less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q3/5

0 of 200 runs suggests the observed result is

A. very unlikely under the model

B. certain

C. impossible

D. typical

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

2552

QUESTION ID

S69-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. very unlikely under the model

GRADING NOTE Rare, not impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 4/200 below 5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

2553

QUESTION ID

S69-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Yes: 2%

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 1

Two-sided comparisons and design choices • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In a caffeine experiment, the observed difference in mean reaction time was 0.04 s, and the question is whether caffeine changes reaction time in either direction. How should the re-randomization results be counted?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

2555

QUESTION ID
S69-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Count differences at least 0.04 s in either direction (|difference| ≥ 0.04)

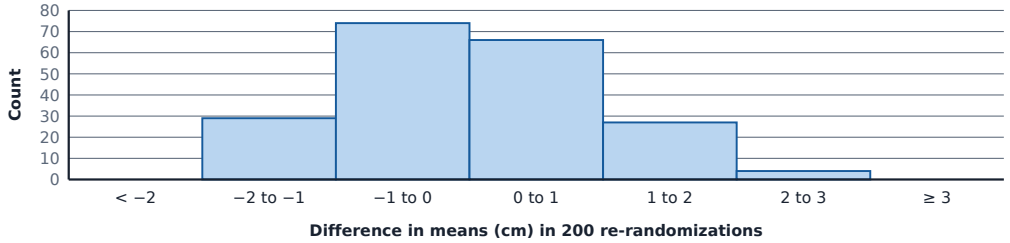
GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using the fertilizer histogram, estimate how many re-randomized differences were 2 or more in absolute value.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

2556

QUESTION ID
S69-F1-Q2

TYPE
Short answer

CORRECT ANSWER
About 4 (no bar below −2, 4 in 2 to 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q2/5

True or false?

Using more plants in each group usually makes the re-randomized differences less spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

2557

QUESTION ID
S69-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Less chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q3/5

A small observed difference can be significant when

A. the groups have very little natural variation

B. the data are rounded

C. the groups were not randomized

D. the sample is small

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

2558

QUESTION ID

S69-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. the groups have very little natural variation

GRADING NOTE Small spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S69-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why should the decision rule (such as 5%) be chosen before seeing the data?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

2559

QUESTION ID

S69-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Choosing it afterward could bias the conclusion toward the result one wants.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 69 • FORWARD 1 • Q5/5

SET 69 • HSS-IC.B.5

FORWARD 2

Evaluate reports of experiments • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A company reports its supplement "improved memory scores by 3 points" in a study of 8 people without a control group. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

2561

QUESTION ID
S69-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Weak evidence: no control group, very small sample, no randomization, no significance test

GRADING NOTE Any two valid criticisms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says a new teaching method "significantly" raised scores in a randomized experiment, with 2 of 200 re-randomizations as extreme. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

2562

QUESTION ID
S69-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Supported: randomized, and 1% is unusual

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q2/5

True or false?

"Statistically significant" means the effect is large and important.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

2563

QUESTION ID
S69-F2-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It means unlikely by chance; it may be small.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which report is most convincing?

A. a study with no comparison group

B. testimonials from customers

C. a randomized experiment with a control group and a small simulated p-value

D. an online poll of users

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

2564

QUESTION ID

S69-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a randomized experiment with a control group and a small simulated p-value

GRADING NOTE Design and evidence.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q4/5

What information should a report of an experiment include?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

2565

QUESTION ID
S69-F2-Q5

TYPE
Short answer

CORRECT ANSWER
How subjects were assigned, the group sizes, the observed difference, and how unusual it is by chance

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S69-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 69 • FORWARD 2 • Q5/5

SET 70 • HSS-IC.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Evaluate claims in reports based on data

MAIN	Algebra II main question	**HSS-IC.B.6**
BACKWARD 1	Random samples and bias	**HSS-IC.A.1**
BACKWARD 2	Study design and causation	**HSS-IC.B.3**
BACKWARD 3	Read scales on graphs	**3.MD.B.3**
FORWARD 1	Rewrite claims so that they follow from the data	**HSS-IC.B.6**
FORWARD 2	Analyze decisions based on test results	**HSS-MD.B.7**

SET 70 • HSS-IC.B.6

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

An ad says "4 out of 5 dentists recommend our toothpaste." Only 5 dentists, chosen by the company, were asked. Evaluate the claim.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

2568

QUESTION ID

S70-M-Q1

TYPE

Short answer

CORRECT ANSWER

Weak: the sample is tiny and not random, so it cannot represent all dentists

GRADING NOTE Form required: evaluation with at least one valid reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "Students who eat breakfast earn higher grades, so breakfast causes higher grades." The data came from a survey. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

2569

QUESTION ID

S70-M-Q2

TYPE

Short answer

CORRECT ANSWER

Not justified: an observational survey shows association, not causation; lurking variables (such as family routines) could explain it

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q2/5

A bar graph shows sales of 50 and 52 thousand, with the vertical axis starting at 48 thousand, so one bar looks twice as tall. Evaluate the impression the graph gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

2570

QUESTION ID
S70-M-Q3

TYPE
Short answer

CORRECT ANSWER
Misleading: the truncated axis exaggerates a 4% difference to look like 100%

GRADING NOTE Form required: evaluation and the actual size of the difference.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A TV show asks viewers to text their vote; 90% of 10,000 texts favor a policy. The show claims most citizens support it. Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

2571

QUESTION ID

S70-M-Q4

TYPE

Short answer

CORRECT ANSWER

Not justified: voluntary response; texters are not representative, regardless of the large number

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-M-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A poll shows candidate A at 51% and B at 49% with a margin of error of ±4%. A headline says "A is winning." Evaluate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

2572

QUESTION ID
S70-M-Q5

TYPE
Short answer

CORRECT ANSWER
Not justified: the difference is well within the margin of error

GRADING NOTE Form required: evaluation and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • MAIN • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 1

Random samples and bias • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Why is a voluntary-response sample biased?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

2574

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

People with strong opinions are more likely to respond

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B1-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q1/5

What does a random sample allow?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

2575

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Generalizing from the sample to the population

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-B1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A very large sample cannot be biased.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q3/5

2576

QUESTION ID

S70-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Size does not fix bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which sample is most likely biased?

A. customers who chose to write reviews

B. a random sample of 1,000

C. randomly chosen customers

D. every customer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q4/5

2577

QUESTION ID

S70-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. customers who chose to write reviews

GRADING NOTE Self-selected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is "5 dentists chosen by the company" a random sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

2578

QUESTION ID

S70-B1-Q5

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 2

Study design and causation • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

ANSWER KEY Reduced copy of the question page, with its answer

What type of study is a survey of students' breakfast habits and grades?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

2580

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

An observational study (sample survey)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What study design could test whether breakfast causes higher grades?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

2581

QUESTION ID
S70-B2-Q2

TYPE
Short answer

CORRECT ANSWER
A randomized experiment

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q2/5

True or false?

Only randomized experiments can establish cause and effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

2582

QUESTION ID

S70-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Random assignment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A confounding variable is

A. the response variable

B. a variable related to both the explanatory and the response variable

C. the sample size

D. the margin of error

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

2583

QUESTION ID
S70-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. a variable related to both the explanatory and the response variable

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name a possible confounder for breakfast and grades.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

2584

QUESTION ID

S70-B2-Q5

TYPE

Short answer

CORRECT ANSWER

For example sleep or family support

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • HSS-IC.B.6

BACKWARD 3

Read scales on graphs • Grade 3 • Measurement & Data • 3.MD.B.3

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph axis starts at 48. Bars reach 50 and 52. What are the bar values?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

2586

QUESTION ID

S70-B3-Q1

TYPE

Short answer

CORRECT ANSWER

50 and 52

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What percent larger is 52 than 50?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

2587

QUESTION ID

S70-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q2/5

True or false?

An axis that does not start at 0 can make differences look larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

2588

QUESTION ID

S70-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Truncation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

On a graph whose axis starts at 48, a bar of 52 looks how many times as tall as a bar of 50?

A. the same

B. 1.04 times

C. 2 times

D. 4 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

2589

QUESTION ID

S70-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2 times

GRADING NOTE Heights 4 and 2 above the axis start.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A bar graph of sales of 50 and 52 (thousand) has a vertical axis starting at 48. How should the graph be redrawn to be fair?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

2590

QUESTION ID

S70-B3-Q5

TYPE

Short answer

CORRECT ANSWER

Start the axis at 0 (or clearly mark the break)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-B3-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 70 • BACKWARD 3 • Q5/5

2590

SET 70 • HSS-IC.B.6

FORWARD 1

Rewrite claims so that they follow from the data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A survey found that students who eat breakfast earn higher grades, and a report says breakfast causes higher grades. Rewrite the claim so that it is supported by the survey.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

2592

QUESTION ID

S70-F1-Q1

TYPE

Short answer

CORRECT ANSWER

"Students who eat breakfast tend to have higher grades" (an association)

GRADING NOTE Any equivalent wording without causation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite the poll headline (51% vs 49%, ±4%) so it is accurate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

2593

QUESTION ID
S70-F1-Q2

TYPE
Short answer

CORRECT ANSWER
"The race is too close to call"

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q2**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q2/5

True or false?

Saying "is associated with" instead of "causes" is appropriate for an observational study.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

2594

QUESTION ID

S70-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Careful wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which conclusion is justified by a random sample of 1,000 voters showing 62% ± 3% support?

A. the sample was biased

B. most voters support the measure

C. exactly 62% support it

D. the measure will pass by 24 points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

2595

QUESTION ID

S70-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. most voters support the measure

GRADING NOTE 59% to 65%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S70-F1-Q4**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An ad says "4 out of 5 dentists recommend our toothpaste," based on 5 dentists chosen by the company. What would the claim need to be convincing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

2596

QUESTION ID

S70-F1-Q5

TYPE

Short answer

CORRECT ANSWER

A large random sample of dentists (and the question wording reported)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 70 • FORWARD 1 • Q5/5

SET 70 • HSS-IC.B.6

FORWARD 2

Analyze decisions based on test results • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A screening test is 95% accurate for people with a disease and gives false positives 5% of the time. If 1% of 10,000 people have the disease, how many positive results are true positives?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

2598

QUESTION ID
S70-F2-Q1

TYPE
Short answer

CORRECT ANSWER
95 true positives out of 95 + 495 = 590 positives

GRADING NOTE Form required: both counts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A report says "a positive test means you almost surely have the disease." Evaluate using your counts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

2599

QUESTION ID
S70-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Misleading: only 95/590 ≈ 16% of positives have the disease

GRADING NOTE Form required: evaluation and probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The probability of having the disease given a positive result can be low even with an accurate test.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

2600

QUESTION ID
S70-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Rare disease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q3/5

2600

ANSWER KEY Reduced copy of the question page, with its answer

With 1% prevalence, P(disease | positive) is about

A. 0.99

B. 0.05

C. 0.95

D. 0.16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

2601

QUESTION ID

S70-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0.16

GRADING NOTE 95/590.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q4/5

What follow-up decision makes sense after a positive screening result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

2602

QUESTION ID
S70-F2-Q5

TYPE
Short answer

CORRECT ANSWER
Confirm with a second, more accurate test before treatment

GRADING NOTE Any reasonable answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S70-F2-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 70 • FORWARD 2 • Q5/5

SET 71 • HSS-MD.B.6 • STATISTICS, INFERENCE, AND DECISIONS

Design and judge fair decision procedures with probabilities

MAIN	Algebra II main question	**HSS-MD.B.6**
BACKWARD 1	Probabilities of equally likely outcomes	**7.SP.C.7.a**
BACKWARD 2	Sample spaces for compound events	**7.SP.C.8.b**
BACKWARD 3	Probability between 0 and 1 and redoing outcomes	**7.SP.C.5**
FORWARD 1	Design fair procedures and count expected trials	**HSS-MD.B.6**
FORWARD 2	Analyze strategies and decisions with probability	**HSS-MD.B.7**

SET 71 • HSS-MD.B.6

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Three friends choose a driver by flipping a coin twice: HH = Ana, HT = Ben, TH = Cy, TT = flip again. Is this fair? Justify with probabilities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

2605

QUESTION ID
S71-M-Q1

TYPE
Short answer

CORRECT ANSWER
Yes: HH, HT, TH are equally likely and TT is redone, so each friend has probability 1/3

GRADING NOTE Form required: decision and probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q1/5

Four people use a die: 1 = A, 2 = B, 3 = C, 4 = D, and 5 or 6 means roll again. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

2606

QUESTION ID
S71-M-Q2

TYPE
Short answer

CORRECT ANSWER
Yes: each person has probability 1/4 after rerolls

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A hat holds 3 slips with Ana's name and 2 with Ben's. One slip is drawn to choose who goes first. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

2607

QUESTION ID
S71-M-Q3

TYPE
Short answer

CORRECT ANSWER
No: P(Ana) = 3/5 and P(Ben) = 2/5

GRADING NOTE Form required: decision and probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A random number generator gives a whole number 1-100. Numbers 1-33 choose A, 34-66 choose B, 67-99 choose C, and 100 means generate again. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

2608

QUESTION ID

S71-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes: each person has 33 equally likely numbers, and 100 is redone

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q4**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q4/5

Two dice are rolled; if the sum is even, player 1 wins, otherwise player 2 wins. Is this fair? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

2609

QUESTION ID
S71-M-Q5

TYPE
Short answer

CORRECT ANSWER
Yes: 18 of the 36 outcomes have an even sum, so each player has probability 1/2

GRADING NOTE Form required: decision and probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6 • Set 71 • MAIN • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 1

Probabilities of equally likely outcomes • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A fair die is rolled. What is P(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

2611

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 3 red and 2 blue slips. What is P(red)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

2612

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Each face of a fair die has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

2613

QUESTION ID

S71-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Uniform model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins: P(HT) is

A. 1/2

B. 1/4

C. 1/8

D. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

2614

QUESTION ID

S71-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/4

GRADING NOTE 4 outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner has 4 equal sections. What is P(one section)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

2615

QUESTION ID

S71-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 71 • BACKWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 2

Sample spaces for compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

List the outcomes of flipping a coin twice.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

2617

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

HH, HT, TH, TT

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How many outcomes does rolling two dice have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

2618

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q2/5

True or false?

There are 3 outcomes for two coin flips: two heads, one head, no heads.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

2619

QUESTION ID

S71-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE These are not equally likely; there are 4 equally likely outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many outcomes for two dice give an even sum?

A. 21

B. 12

C. 18

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

2620

QUESTION ID

S71-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 18

GRADING NOTE Half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the outcomes of rolling two dice with a sum of 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

2621

QUESTION ID

S71-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(1, 3), (2, 2), (3, 1)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 71 • BACKWARD 2 • Q5/5

SET 71 • HSS-MD.B.6

BACKWARD 3

Probability between 0 and 1 and redoing outcomes • Grade 7 • Statistics & Probability • 7.SP.C.5

ANSWER KEY Reduced copy of the question page, with its answer

If an outcome is ignored and the experiment is redone, how does that affect the remaining outcomes' chances?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

2623

QUESTION ID

S71-B3-Q1

TYPE

Short answer

CORRECT ANSWER

They stay equally likely to each other and their probabilities are scaled up to add to 1

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

With rerolls on 5 and 6, what is P(1) for a die?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

2624

QUESTION ID

S71-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q2/5

True or false?

Probabilities of all outcomes in a sample space add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

2625

QUESTION ID

S71-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Total probability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

An event that can never happen has probability

A. 1

B. −1

C. 1/2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

2626

QUESTION ID

S71-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE Impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Three equally likely outcomes: what is each probability?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

2627

QUESTION ID

S71-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-B3-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 71 • BACKWARD 3 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 1

Design fair procedures and count expected trials • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Design a fair way to choose one of 5 people using a fair coin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

2629

QUESTION ID
S71-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Flip 3 times (8 outcomes); assign 5 outcomes to the people and redo the other 3

GRADING NOTE Any valid design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Three friends choose a driver by flipping a coin twice: HH, HT, and TH each pick a friend, and TT means flip again. What is the probability that a choice is made on the first try?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

2630

QUESTION ID
S71-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A fair procedure must give each person an equal chance, even if it may take several tries.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

2631

QUESTION ID
S71-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Fairness is about probabilities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

To choose among 6 people fairly with one die, you

A. roll twice and add

B. assign one face to each person

C. assign 1-3 to person A

D. reroll on 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

2632

QUESTION ID
S71-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. assign one face to each person

GRADING NOTE Six equally likely faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A teacher picks a student from a class of 28 with a random number generator 1-30. Design a fair rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

2633

QUESTION ID
S71-F1-Q5

TYPE
Short answer

CORRECT ANSWER
Assign numbers 1-28 to students and regenerate on 29 or 30

GRADING NOTE Any valid design.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F1-Q5
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 71 • FORWARD 1 • Q5/5

SET 71 • HSS-MD.B.6

FORWARD 2

Analyze strategies and decisions with probability • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A game costs $1. You roll a die and win $4 if you roll a 6. Find the expected net gain per game.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

2635

QUESTION ID
S71-F2-Q1

TYPE
Short answer

CORRECT ANSWER
4/6 − 1 = −1/3 dollar (lose about $0.33 per game)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q1
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A game costs $1; you roll a die and win $4 if you roll a 6. Is the game fair in the sense of expected value? What prize would make it fair?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

2636

QUESTION ID
S71-F2-Q2

TYPE
Short answer

CORRECT ANSWER
No; a $6 prize makes the expected net gain 0

GRADING NOTE Form required: decision and prize.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q2/5

True or false?

A game can be fair in who is chosen but still have a negative expected value for the player.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

2637

QUESTION ID
S71-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Two meanings of fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

With P(win) = 1/4 and prize \$8 for a \$2 ticket, the expected net gain is

A. \$2

B. −\$2

C. \$0

D. \$6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

2638

QUESTION ID
S71-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. \$0

GRADING NOTE 8/4 − 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S71-F2-Q4**
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain the difference between a fair choice procedure and a fair game.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

2639

QUESTION ID
S71-F2-Q5

TYPE
Short answer

CORRECT ANSWER
A fair choice gives everyone equal probability; a fair game has expected net gain 0.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S71-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 71 • FORWARD 2 • Q5/5

SET 72 • HSS-MD.B.7 • STATISTICS, INFERENCE, AND DECISIONS

Analyze decisions and strategies with probabilities

MAIN	Algebra II main question	**HSS-MD.B.7**
BACKWARD 1	Probability of compound independent events	**7.SP.C.8.a**
BACKWARD 2	Two-way tables of counts	**8.SP.A.4**
BACKWARD 3	Weighted averages of outcomes	**6.RP.A.3.c**
FORWARD 1	Investigate how changing conditions changes the decision	**HSS-MD.B.7**
FORWARD 2	Find expected values of random variables	**HSS-MD.A.2**

SET 72 • HSS-MD.B.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A disease affects 1% of 10,000 people. A test detects 95% of cases and gives false positives for 5% of healthy people. Find P(disease | positive).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

2642

QUESTION ID

S72-M-Q1

TYPE

Short answer

CORRECT ANSWER

95/590 ≈ 0.161

GRADING NOTE Form required: the probability with the counts. 95 true positives, 495 false positives.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q1**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A shipment is accepted if 2 randomly tested items are both non-defective. If 10% of items are defective, what is the probability the shipment is accepted? (Assume independence.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

2643

QUESTION ID

S72-M-Q2

TYPE

Short answer

CORRECT ANSWER

$0.9 \times 0.9 = 0.81$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q2**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A $30 warranty covers a repair that costs $200 and happens with probability 0.1. Compare the expected repair cost without the warranty to the warranty price, and decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

2644

QUESTION ID

S72-M-Q3

TYPE

Short answer

CORRECT ANSWER

Expected cost
0.1 × 200
= $20 < $30, so on average the warranty is not worth buying

GRADING NOTE Form required: expected cost, comparison, decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q3/5

Option A pays $10 with probability 0.5; option B pays $50 with probability 0.1. Compare the expected values and the risks.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

2645

QUESTION ID

S72-M-Q4

TYPE

Short answer

CORRECT ANSWER

Both have expected value $5; B is riskier (usually pays nothing)

GRADING NOTE Form required: both expected values and a risk comparison.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q4/5

A test gives +1 point for a correct answer and −1/4 point for a wrong one on 4-choice questions. Should a student guess randomly on a question? Justify with expected value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

2646

QUESTION ID

S72-M-Q5

TYPE

Short answer

CORRECT ANSWER

Yes (slightly): expected value = 1/4(1) + 3/4(−1/4) = 1/16 > 0

GRADING NOTE Form required: expected value and decision.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7 • Set 72 • MAIN • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 1

Probability of compound independent events • Grade 7 • Statistics & Probability • 7.SP.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

P(non-defective) = 0.9 for each item. Find P(two non-defective).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

2648

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.81

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two fair coins: P(both heads)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

2649

QUESTION ID

S72-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

For independent events, P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

2650

QUESTION ID

S72-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Multiplication rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

P(three non-defective) with P = 0.9 each is

A. 0.9

B. 0.27

C. 0.729

D. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

2651

QUESTION ID

S72-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.729

GRADING NOTE 0.9^3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find P(at least one defective) among two items when P(defective) = 0.1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

2652

GRADING NOTE Any equivalent form. 1 − 0.81.

QUESTION ID

S72-B1-Q5

TYPE

Short answer

CORRECT ANSWER

0.19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 72 • BACKWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 2

Two-way tables of counts • Grade 8 • Statistics & Probability • 8.SP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Of 10,000 people, 1% have a disease. How many have it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

2654

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Of 9,900 healthy people, 5% test positive. How many?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

2655

QUESTION ID

S72-B2-Q2

TYPE

Short answer

CORRECT ANSWER

495

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q2/5

True or false?

A two-way table can show test results for people with and without a disease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

2656

QUESTION ID

S72-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rows and columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If 95 sick and 495 healthy people test positive, the total positives are

A. 10,000

B. 590

C. 95

D. 495

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

2657

QUESTION ID

S72-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 590

GRADING NOTE Add.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What fraction of positives are sick (95 of 590)? Give a decimal to the nearest thousandth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

2658

QUESTION ID

S72-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0.161

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • HSS-MD.B.7

BACKWARD 3

Weighted averages of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 10% of $200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

2660

QUESTION ID

S72-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.5 × 10 and 0.1 × 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

2661

QUESTION ID

S72-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5 and 5

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1/4 \times 1 + 3/4 \times (-1/4) = 1/4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q3/5

2662

QUESTION ID

S72-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1/4 - 3/16 = 1/16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

0.2 × 30 + 0.8 × 0 equals

A. 30

B. 24

C. 0.2

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

2663

QUESTION ID

S72-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6

GRADING NOTE Weighted sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute 0.9 × 0 + 0.1 × 200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

2664

QUESTION ID

S72-B3-Q5

TYPE

Short answer

CORRECT ANSWER

20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 72 • BACKWARD 3 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 1

Investigate how changing conditions changes the decision • High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A test detects 95% of cases and gives false positives for 5% of healthy people. In a group of 10,000 where 10% have the disease, find P(disease | positive).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

2666

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

950/1400 ≈ 0.679

GRADING NOTE Form required: the probability. 950 true and 450 false positives in 10,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A $30 warranty covers a repair that costs $200. If the repair probability is 0.2, is the warranty worth it on average?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

2667

QUESTION ID
S72-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes: expected cost $40 > $30

GRADING NOTE Form required: decision and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q2/5

True or false?

A positive result is more convincing when the disease is more common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

2668

QUESTION ID
S72-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Compare 0.161 and 0.679.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q3
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q3/5

If students could eliminate one wrong choice before guessing (3 choices left), the expected value of guessing is

A. $1/3 - 2/3 \cdot 1/4 = 1/6$

B. 0

C. $1/16$

D. $-1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

2669

QUESTION ID
S72-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $1/3 - 2/3 \cdot 1/4 = 1/6$

GRADING NOTE Higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q4
High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

How many items must be tested (all non-defective) for the acceptance probability to drop below 0.5 when 10% are defective?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

2670

QUESTION ID
S72-F1-Q5

TYPE
Short answer

CORRECT ANSWER
7 items ($0.9^7 \approx 0.478$)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 72 • FORWARD 1 • Q5/5

SET 72 • HSS-MD.B.7

FORWARD 2

Find expected values of random variables • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A random variable X takes values 0, 1, 2 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

2672

QUESTION ID

S72-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A raffle sells 500 tickets at $2; one prize is $400. Find the expected net gain of one ticket.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

2673

QUESTION ID
S72-F2-Q2

TYPE
Short answer

CORRECT ANSWER
400/500 − 2 = −$1.20

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q2/5

True or false?

The expected value is the long-run average of many repetitions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

2674

QUESTION ID
S72-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q3
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For a fair die, E(roll) is

A. 3

B. 21

C. 3.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

2675

QUESTION ID
S72-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 3.5

GRADING NOTE 21/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S72-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q4/5

A $30 warranty covers a $200 repair that happens with probability 0.1. Explain how deciding whether to buy it is an expected-value calculation for a random variable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

2676

QUESTION ID
S72-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The repair cost is a random variable (0 or 200); its expected value 20 is compared with the certain cost 30.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S72-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 72 • FORWARD 2 • Q5/5

SET 73 • HSN-CN.A.3 • ADVANCED EXTENSION (+)

Use conjugates to find moduli and quotients of complex numbers

MAIN	Algebra II main question	**HSN-CN.A.3**
BACKWARD 1	Multiply complex numbers	**HSN-CN.A.2**
BACKWARD 2	Distance from the origin	**8.G.B.8**
BACKWARD 3	Powers of i	**HSN-CN.A.1**
FORWARD 1	Prove properties of conjugates and moduli	**HSN-CN.A.3**
FORWARD 2	Write complex numbers in polar form	**HSN-CN.B.4**

SET 73 • HSN-CN.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Give the conjugate of $3 - 4i$ and find $|3 - 4i|$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

2679

QUESTION ID

S73-M-Q1

TYPE

Short answer

CORRECT ANSWER

Conjugate $3 + 4i$; $|3 - 4i| = 5$

GRADING NOTE Form required: both. $\sqrt{(9 + 16)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (2 + 3i)/(1 − i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

2680

QUESTION ID

S73-M-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2 + (5/2)i

GRADING NOTE Form required: a + bi. Multiply by (1 + i)/(1 + i).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write (5 + i)/(2 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

2681

QUESTION ID

S73-M-Q3

TYPE

Short answer

CORRECT ANSWER

1 − i

GRADING NOTE Form required: a + bi. (5 + i)(2 − 3i)/13 = (13 − 13i)/13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $|-5 + 12i|$ and compute $(-5 + 12i)(-5 - 12i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q4/5

2682

QUESTION ID
S73-M-Q4

TYPE
Short answer

CORRECT ANSWER
13 and 169

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q4**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1/(4 + 3i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-M-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

2683

QUESTION ID

S73-M-Q5

TYPE

Short answer

CORRECT ANSWER

4/25 − (3/25)i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-M-Q5**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • MAIN • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 1

Multiply complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Multiply (2 + 3i)(1 + i).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

2685

QUESTION ID

S73-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−1 + 5i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(3 - 4i)(3 + 4i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

2686

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q2/5

True or false?

$(a + bi)(a - bi) = a^2 + b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q3/5

2687

QUESTION ID

S73-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $-b^2i^2 = b^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(5 + i)(2 − 3i) equals

A. 13 + 13i

B. 7 − 13i

C. 10 − 3i

D. 13 − 13i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

2688

QUESTION ID

S73-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13 − 13i

GRADING NOTE 10 − 15i + 2i + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply $(1 - i)(1 + i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

2689

QUESTION ID

S73-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 2

Distance from the origin • Grade 8 • Geometry • 8.G.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (3, −4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

2691

QUESTION ID

S73-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (−5, 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

2692

QUESTION ID

S73-B2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The distance from (0, 0) to (a, b) is $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

2693

QUESTION ID

S73-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Pythagorean Theorem.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q3/5

2693

The distance from (0, 0) to (1, 1) is

A. 0

B. 1

C. 2

D. √2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

2694

QUESTION ID

S73-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. √2

GRADING NOTE √(1 + 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance from (0, 0) to (4, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

2695

QUESTION ID

S73-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • HSN-CN.A.3

BACKWARD 3

Powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

2697

QUESTION ID

S73-B3-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-12i \cdot i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

2698

QUESTION ID

S73-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q2/5

True or false?

$-3i^2 = -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

2699

QUESTION ID

S73-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^2 = -1$, so $-3i^2 = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is real?

A. 3i

B. $(4 + 3i)^2$

C. $(4 + 3i)(4 - 3i)$

D. $4 + 3i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

2700

QUESTION ID

S73-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(4 + 3i)(4 - 3i)$

GRADING NOTE Conjugate product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 13/13 − 13i/13 in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

2701

QUESTION ID

S73-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1 − i

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 1

Prove properties of conjugates and moduli • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which argument proves $z \cdot \bar{z} = |z|^2$ for $z = a + bi$ (a and b real)?

A. $(a + bi)(a - bi) = a^2 + b^2i^2$, which is $|z|^2$ because $|z| = \sqrt{(a^2 + b^2i^2)}$.

B. $(a + bi)(a - bi) = a^2 - b^2$, and $|z|^2 = a^2 - b^2$.

C. $(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

D. For $z = 3 + 4i$: $(3 + 4i)(3 - 4i) = 25 = 5^2$, so the statement holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

2703

QUESTION ID

S73-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C.

$(a + bi)(a - bi) = a^2 - b^2i^2 = a^2 + b^2$, and $|z|^2 = a^2 + b^2$.

GRADING NOTE Multiply, use $i^2 = -1$, and compare with $|z| = \sqrt{(a^2 + b^2)}$. The others forget $i^2 = -1$ (and misstate $|z|$), check one example, or expand with a sign error.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Verify |zw| = |z||w| for z = 3 + 4i and w = 1 + i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

2704

QUESTION ID

S73-F1-Q2

TYPE

Short answer

CORRECT ANSWER

zw = −1 + 7i,
|zw| = √50 = 5√2 = 5 · √2

GRADING NOTE Form required: both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q2/5

True or false?

The conjugate of a real number is itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

2705

QUESTION ID

S73-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE b = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $z = 2 - 5i$, then $z + \bar{z}$ is

A. $4 - 10i$

B. 0

C. $-10i$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

2706

QUESTION ID

S73-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 4

GRADING NOTE 2a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\bar{z}/|z|^2$ to write $1/z$ for $z = 1 + 2i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

2707

QUESTION ID

S73-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$(1 - 2i)/5 = 1/5 - (2/5)i$

GRADING NOTE Form required: a + bi.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F1-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 73 • FORWARD 1 • Q5/5

SET 73 • HSN-CN.A.3

FORWARD 2

Write complex numbers in polar form • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Write 1 + i in polar form r(cos θ + i sin θ).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

2709

QUESTION ID

S73-F2-Q1

TYPE

Short answer

CORRECT ANSWER

√2(cos(π/4) + i sin(π/4))

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 2(cos(π/2) + i sin(π/2)) in rectangular form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

2710

QUESTION ID
S73-F2-Q2

TYPE
Short answer

CORRECT ANSWER
2i

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q2/5

True or false?

The modulus of a complex number is the r in its polar form.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

2711

QUESTION ID
S73-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S73-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The polar form of -3 is

A. $3(\cos \pi + i \sin \pi)$

B. $-3(\cos \pi + i \sin \pi)$

C. $3(\cos 0 + i \sin 0)$

D. $3(\cos(\pi/2) + i \sin(\pi/2))$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

2712

QUESTION ID

S73-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3(cos π + i sin π)

GRADING NOTE Angle π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain what the modulus and the angle of a complex number represent in the complex plane.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

2713

QUESTION ID
S73-F2-Q5

TYPE
Short answer

CORRECT ANSWER
The modulus is the distance from 0; the angle is the direction measured from the positive real axis.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S73-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 73 • FORWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Graph rational functions with zeros and asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Rewrite a rational expression by inspection	**HSA-APR.D.6**
BACKWARD 2	Domain and excluded values	**HSF-IF.B.5**
BACKWARD 3	Sign of a quotient on intervals	**7.NS.A.2**
FORWARD 1	Graph rational functions with slant asymptotes	**HSF-IF.C.7.d**
FORWARD 2	Interpret a rational model's features in context	**HSF-IF.B.4**

SET 74 • HSF-IF.C.7.d

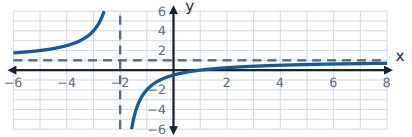
MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

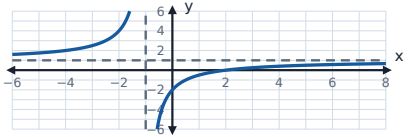
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = (x + 1)/(x − 2), with its asymptotes dashed?

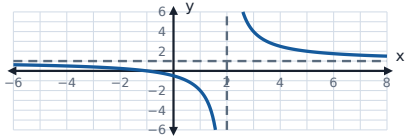
A



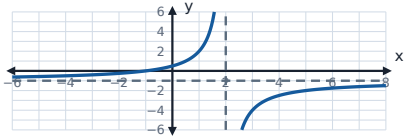
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

2716

QUESTION ID

S74-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Zero −1, y-intercept −1/2, vertical asymptote x = 2, horizontal asymptote y = 1 (ratio of leading coefficients); $y \to +\infty$ as $x \to 2^+$ and $y \to -\infty$ as $x \to 2^-$. The others show (x − 1)/(x + 2) (same y-intercept), (x − 2)/(x + 1), and the reflection.

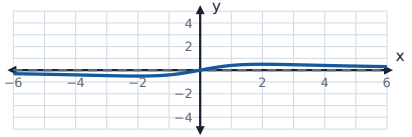
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q1/5

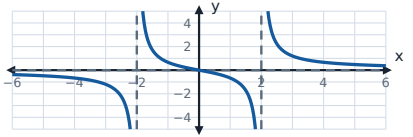
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2x/(x^2 - 4)$, with its asymptotes dashed?

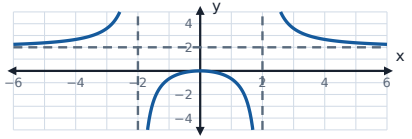
A



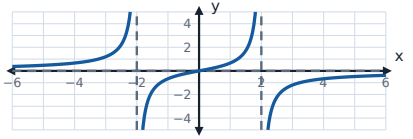
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

2717

QUESTION ID

S74-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Zero and y-intercept 0; vertical asymptotes $x = \pm 2$ (zeros of $x^2 - 4$); horizontal asymptote $y = 0$ (degree of numerator < degree of denominator); e.g. $y(3) = 6/5 > 0$ and $y(1) = -2/3 < 0$. The others show $2x/(x^2 + 4)$, $2x^2/(x^2 - 4)$, and the reflection.

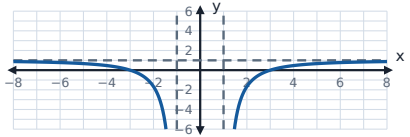
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q2/5

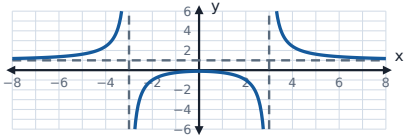
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 - 1)/(x^2 - 9)$, with its asymptotes dashed?

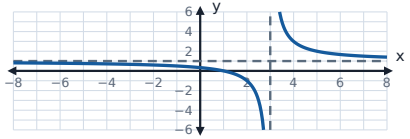
A



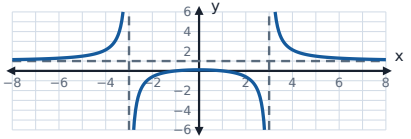
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

2718

QUESTION ID

S74-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Zeros ±1, y-intercept 1/9, vertical asymptotes $x = \pm 3$, horizontal asymptote $y = 1$; the outer branches are above $y = 1$. The others show the reciprocal $(x^2 - 9)/(x^2 - 1)$, $(x^2 + 1)/(x^2 - 9)$, and $(x - 1)/(x - 3)$.

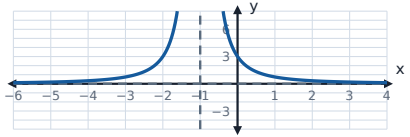
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q3/5

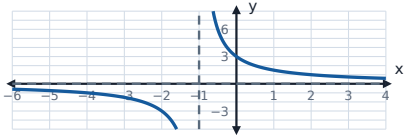
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 3/(x + 1)^2$, with its asymptotes dashed?

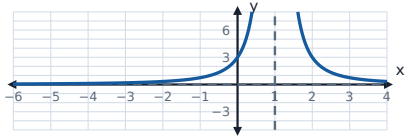
A



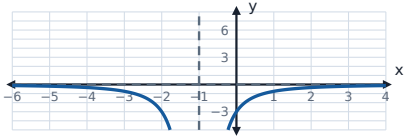
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

2719

QUESTION ID

S74-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE No zeros; y-intercept 3; vertical asymptote $x = -1$ with $y \to +\infty$ on both sides (the square is positive); horizontal asymptote $y = 0$. The others show $3/(x + 1)$, $3/(x - 1)^2$ (same y-intercept), and $-3/(x + 1)^2$.

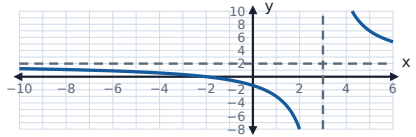
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q4/5

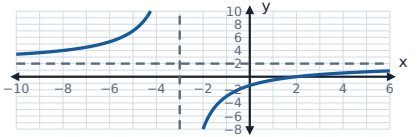
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = (2x − 4)/(x + 3), with its asymptotes dashed?

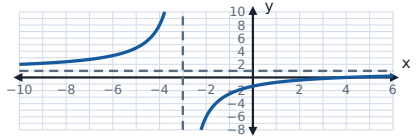
A



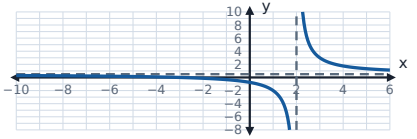
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

2720

QUESTION ID

S74-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Zero 2, y-intercept −4/3, vertical asymptote x = −3, horizontal asymptote y = 2/1 = 2. The others show (2x + 4)/(x − 3) and (x − 4)/(x + 3) (both with the same y-intercept) and the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • MAIN • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 1

Rewrite a rational expression by inspection • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite (x + 1)/(x − 2) as 1 + r/(x − 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

2722

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1 + 3/(x − 2)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-B1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $(2x - 4)/(x + 3)$ as $2 + r/(x + 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

2723

QUESTION ID
S74-B1-Q2

TYPE
Short answer

CORRECT ANSWER
2 − 10/(x + 3)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q2/5

True or false?

The constant q in q + r/(x − a) is the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

2724

QUESTION ID
S74-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE r/(x − a) → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x^2 - 1)/(x^2 - 9)$ equals

A. $1 + 8/(x^2 - 9)$

B. $1 - 8/(x^2 - 9)$

C. $1 + 1/(x^2 - 9)$

D. $1/9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

2725

QUESTION ID

S74-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $1 + 8/(x^2 - 9)$

GRADING NOTE $x^2 - 1 = (x^2 - 9) + 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor the denominator of $2x/(x^2 - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

2726

QUESTION ID

S74-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x - 2)(x + 2)$

GRADING NOTE Form required: factored. The vertical asymptotes are at the zeros of these factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-B1-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 2

Domain and excluded values • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $(x + 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

2728

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \neq 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $3/(x + 1)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

2729

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \neq -1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q2/5

True or false?

A rational function is undefined where its denominator is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

2730

QUESTION ID

S74-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Division by 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q3/5

2730

ANSWER KEY Reduced copy of the question page, with its answer

$2x/(x^2 - 4)$ is undefined at

A. $x = \pm 2$

B. $x = 4$

C. $x = 2$ only

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

2731

QUESTION ID

S74-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x = \pm 2$

GRADING NOTE $x^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

List the excluded values of $(x^2 - 1)/(x^2 - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

2732

QUESTION ID

S74-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3 and −3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.d

BACKWARD 3

Sign of a quotient on intervals • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Is (x + 1)/(x − 2) positive or negative at x = 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

2734

QUESTION ID

S74-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Negative: 1/(−2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $2x/(x^2 - 4)$ positive or negative at $x = 3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

2735

QUESTION ID

S74-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Positive: 6/5

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q2/5

True or false?

A quotient of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

2736

QUESTION ID

S74-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For example $(-6)/(-2) = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

At x = 2.1, (x + 1)/(x − 2) is

A. large and negative

B. undefined

C. zero

D. large and positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

2737

QUESTION ID

S74-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. large and positive

GRADING NOTE 3.1/0.1 = 31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $3/(x + 1)^2$ ever negative? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

2738

QUESTION ID

S74-B3-Q5

TYPE

Short answer

CORRECT ANSWER

No: a positive number over a square

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-B3-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • HSF-IF.C.7.d

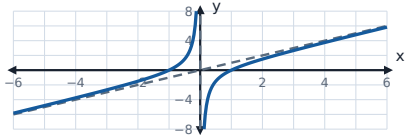
FORWARD 1

Graph rational functions with slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

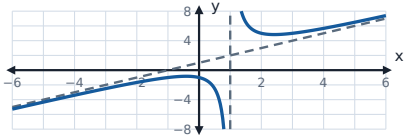
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 + 1)/x$, with its asymptotes dashed?

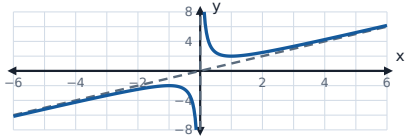
A



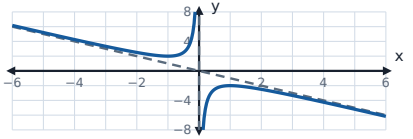
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

2740

QUESTION ID

S74-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $(x^2 + 1)/x = x + 1/x$: vertical asymptote $x = 0$, slant asymptote $y = x$, no zeros, no y-intercept, through $(1, 2)$ and $(-1, -2)$. The others show $(x^2 - 1)/x$, $(x^2 + 1)/(x - 1)$, and $-(x^2 + 1)/x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slant asymptote of $y = (x^2 - x - 1)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

2741

QUESTION ID

S74-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$y = x + 1$

GRADING NOTE Form required: equation. Quotient x + 1, remainder 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q2/5

True or false?

A rational function can cross its horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

2742

QUESTION ID

S74-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example $y = x/(x^2 + 1)$ crosses $y = 0$ at $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = (3x^2 + 1)/(x^2 + 4)$ has horizontal asymptote

A. none

B. y = 3

C. y = 0

D. y = 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

2743

QUESTION ID

S74-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y = 3

GRADING NOTE Ratio of leading coefficients.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a rational function with vertical asymptote x = 1, horizontal asymptote y = 2, and zero at x = 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

2744

QUESTION ID

S74-F1-Q5

TYPE

Short answer

CORRECT ANSWER

For example

2(x − 3)/(x − 1)

GRADING NOTE Any valid function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 74 • FORWARD 1 • Q5/5

SET 74 • HSF-IF.C.7.d

FORWARD 2

Interpret a rational model's features in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The average cost per item is A(x) = (500 + 4x)/x. Interpret the vertical and horizontal asymptotes in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

2746

QUESTION ID

S74-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = 0: average cost is undefined/huge for very few items; y = 4: the average cost approaches $4 per item

GRADING NOTE Form required: both interpretations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A drug concentration is $C(t) = 20t/(t^2 + 4)$. Interpret its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

2747

QUESTION ID
S74-F2-Q2

TYPE
Short answer

CORRECT ANSWER
y = 0: the concentration eventually approaches 0

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q2/5

True or false?

For a context, only the part of a rational graph in the context domain is meaningful.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

2748

QUESTION ID

S74-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In A(x) = (500 + 4x)/x, as x grows the average cost

A. increases

B. is constant

C. approaches 0

D. decreases toward 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

2749

QUESTION ID

S74-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. decreases toward 4

GRADING NOTE 500/x → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the number of items for which the average cost is $5.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S74-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

2750

QUESTION ID

S74-F2-Q5

TYPE

Short answer

CORRECT ANSWER

500

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 74 • FORWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d • ADVANCED EXTENSION (+)

Distinguish holes from vertical asymptotes

MAIN	Algebra II main question	**HSF-IF.C.7.d**
BACKWARD 1	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 2	Simplify rational expressions and keep exclusions	**HSA-APR.D.7**
BACKWARD 3	Evaluate the simplified expression	**HSF-IF.A.2**
FORWARD 1	Construct and graph rational functions with holes	**HSF-IF.C.7.d**
FORWARD 2	Relate holes to extraneous solutions	**HSA-REI.A.2**

SET 75 • HSF-IF.C.7.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

Find every hole and vertical asymptote of $f(x) = (x^2 - 4)/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

2753

QUESTION ID

S75-M-Q1

TYPE

Short answer

CORRECT ANSWER

Hole at (2, 4); no vertical asymptote

GRADING NOTE Form required: both answers. $f(x) = x + 2$ for $x \neq 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every hole and vertical asymptote of $f(x) = (x - 3)/(x^2 - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

2754

QUESTION ID
S75-M-Q2

TYPE
Short answer

CORRECT ANSWER
Hole at (3, 1/6); vertical asymptote x = −3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every hole and vertical asymptote of $f(x) = (x^2 + x - 6)/(x^2 - x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

2755

QUESTION ID
S75-M-Q3

TYPE
Short answer

CORRECT ANSWER
Hole at (2, 5/3); vertical asymptote x = −1

GRADING NOTE Form required: both. (x + 3)(x − 2)/((x − 2)(x + 1)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every hole and vertical asymptote of $f(x) = (x + 1)/(x^2 - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

2756

QUESTION ID

S75-M-Q4

TYPE

Short answer

CORRECT ANSWER

Hole at (−1, −1/2); vertical asymptote x = 1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find every hole and vertical asymptote of $f(x) = x(x - 1)/(x(x + 2))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

2757

QUESTION ID

S75-M-Q5

TYPE

Short answer

CORRECT ANSWER

Hole at (0, −1/2); vertical asymptote x = −2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • MAIN • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 1

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + x - 6$ and $x^2 - x - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

2759

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x − 2) and (x − 2)(x + 1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

2760

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(x − 3)(x + 3)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 - 1 = (x - 1)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

2761

QUESTION ID

S75-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x - 1)^2 = x^2 - 2x + 1$; $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

2762

QUESTION ID

S75-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (x − 2)(x + 2)

GRADING NOTE Difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 + 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

2763

QUESTION ID

S75-B1-Q5

TYPE

Short answer

CORRECT ANSWER

x(x + 2)

GRADING NOTE Form required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 2

Simplify rational expressions and keep exclusions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (x − 3)/((x − 3)(x + 3)) and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q1
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

2765

QUESTION ID
S75-B2-Q1

TYPE
Short answer

CORRECT ANSWER
1/(x + 3); x ≠ ±3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-B2-Q1**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify (x + 1)/((x − 1)(x + 1)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

2766

QUESTION ID
S75-B2-Q2

TYPE
Short answer

CORRECT ANSWER
1/(x − 1), x ≠ ±1

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q2
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

After simplifying, the original excluded values are still excluded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

2767

QUESTION ID
S75-B2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Domain kept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q3
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q3/5

$x(x - 1)/(x(x + 2))$ simplifies to

A. $-1/2$

B. $(x - 1)/(x + 2)$ for all x

C. $x - 1$

D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q4
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

2768

QUESTION ID
S75-B2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $(x - 1)/(x + 2)$, $x \neq 0, -2$

GRADING NOTE Cancel x but keep $x \neq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S75-B2-Q4**
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(x^2 - 4)/(x - 2)$ and state the excluded value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

2769

QUESTION ID
S75-B2-Q5

TYPE
Short answer

CORRECT ANSWER
x + 2, x ≠ 2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B2-Q5
High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • HSF-IF.C.7.d

BACKWARD 3

Evaluate the simplified expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 1/(x + 3) at x = 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

2771

QUESTION ID

S75-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate (x + 3)/(x + 1) at x = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

2772

QUESTION ID

S75-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q2/5

True or false?

The y-coordinate of a hole is the value of the simplified expression there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q3/5

2773

QUESTION ID

S75-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Limit value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x - 1)/(x + 2)$ at $x = 0$ is

A. 1/2

B. −1/2

C. undefined

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

2774

QUESTION ID

S75-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −1/2

GRADING NOTE −1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 1/(x − 1) at x = −1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

2775

QUESTION ID

S75-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 3 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 1

Construct and graph rational functions with holes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Write a rational function with a hole at x = 1, a vertical asymptote x = −3, and horizontal asymptote y = 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

2777

QUESTION ID

S75-F1-Q1

TYPE

Short answer

CORRECT ANSWER

For example (x − 1)/((x − 1)(x + 3))

GRADING NOTE Any valid function.

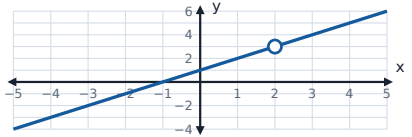
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q1/5

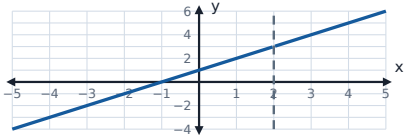
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (x^2 - x - 2)/(x - 2)$?

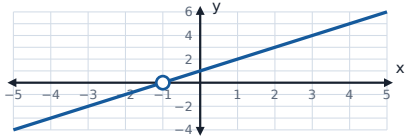
A



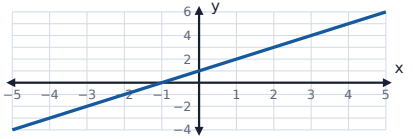
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

2778

QUESTION ID

S75-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $x^2 - x - 2 = (x - 2)(x + 1)$, so $y = x + 1$ for $x \neq 2$: the line with a hole at (2, 3). The factor cancels, so there is no asymptote at $x = 2$, and $x = -1$ is a zero, not an excluded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q2/5

True or false?

A rational function can have both a hole and a vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

2779

QUESTION ID

S75-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Different factors.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 4)^2/(x - 4)$ has

A. no special features

B. a hole at (4, 4)

C. a vertical asymptote $x = 4$

D. a hole at (4, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q4/5

2780

QUESTION ID

S75-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a hole at (4, 0)

GRADING NOTE Simplifies to $x - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $x = -3$ is an asymptote but $x = 3$ is a hole for $(x - 3)/(x^2 - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

2781

QUESTION ID

S75-F1-Q5

TYPE

Short answer

CORRECT ANSWER

After canceling $x - 3$, the factor $x + 3$ remains in the denominator (the values grow without bound near -3), but $x - 3$ does not (values approach 1/6 near 3).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 75 • FORWARD 1 • Q5/5

SET 75 • HSF-IF.C.7.d

FORWARD 2

Relate holes to extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x^2 - 4)/(x - 2) = 4$. Explain the result using the hole.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

2783

QUESTION ID
S75-F2-Q1

TYPE
Short answer

CORRECT ANSWER
No solution: x + 2 = 4 gives x = 2, which is excluded (the hole at (2, 4))

GRADING NOTE Form required: answer and explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x - 3)/(x^2 - 9) = 1/6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

2784

QUESTION ID

S75-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No solution: 1/(x + 3) = 1/6 gives x = 3, which is excluded

GRADING NOTE Form required: answer and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of f(x) = c can occur exactly at a hole whose height is c.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

2785

QUESTION ID

S75-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Graph has no point there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)/(x^2 - 1) = -1/2$ has

A. no solution

B. $x = -1$

C. $x = 0$

D. $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q4/5

2786

QUESTION ID

S75-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. no solution

GRADING NOTE The only candidate is the hole.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $(x^2 + x - 6)/(x^2 - x - 2) = 2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

2787

QUESTION ID

S75-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. (x + 3)/(x + 1) = 2 gives x + 3 = 2x + 2, so x = 1, which is not excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S75-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 75 • FORWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c • ADVANCED EXTENSION (+)

Compose functions and interpret compositions with domains

MAIN	Algebra II main question	**HSF-BF.A.1.c**
BACKWARD 1	Build combined models	**HSF-BF.A.1.b**
BACKWARD 2	Evaluate a function at an expression	**HSF-IF.A.2**
BACKWARD 3	Domains of radical, rational, and log functions	**HSF-IF.B.5**
FORWARD 1	Decompose functions into compositions	**HSF-BF.A.1.c**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 76 • HSF-BF.A.1.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{x}$ and $g(x) = x - 4$. Write $f(g(x))$ and $g(f(x))$ and give the domain of each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

2790

QUESTION ID

S76-M-Q1

TYPE

Short answer

CORRECT ANSWER

$f(g(x)) = \sqrt{(x - 4)}$, $x \geq 4$; $g(f(x)) = \sqrt{x} - 4$, $x \geq 0$

GRADING NOTE Form required: both with domains.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The temperature is T(h) = 20 − 6.5h °C at height h km, and a balloon's height is h(t) = 0.3t km after t minutes. Write T(h(t)) and find the temperature at t = 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

2791

QUESTION ID

S76-M-Q2

TYPE

Short answer

CORRECT ANSWER

T(h(t)) = 20 − 1.95t; 0.5°C

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 1/(x - 1)$ and $g(x) = x^2$. Write $f(g(x))$ and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

2792

QUESTION ID

S76-M-Q3

TYPE

Short answer

CORRECT ANSWER

$1/(x^2 - 1)$; $x \neq 1, -1$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A dollar amount d converts to euros by e(d) = 0.9d, and a store's price with tax in euros is p(e) = 1.2e. Write p(e(d)) and find it for d = 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

2793

QUESTION ID

S76-M-Q4

TYPE

Short answer

CORRECT ANSWER

p(e(d)) = 1.08d; 54 euros

GRADING NOTE Form required: function and value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$ and $g(x) = x^2 - 9$. Write $f(g(x))$ and give its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

2794

QUESTION ID
S76-M-Q5

TYPE
Short answer

CORRECT ANSWER
$\log_2(x^2 - 9)$; $x < -3$ or $x > 3$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • MAIN • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 1

Build combined models • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) + g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

2796

QUESTION ID

S76-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + x + 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x + 1$. Find $f(x) \cdot g(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

2797

QUESTION ID

S76-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 + x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

f(g(x)) is a different combination from f(x) · g(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

2798

QUESTION ID
S76-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Composition vs product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = 2x and g(x) = x + 3, f(x) − g(x) is

A. x − 3

B. $2x^2 + 6x$

C. x + 3

D. 3x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

2799

QUESTION ID

S76-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x − 3

GRADING NOTE Subtract.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name the operation in T(h(t)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

2800

QUESTION ID

S76-B1-Q5

TYPE

Short answer

CORRECT ANSWER

Composition

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 2

Evaluate a function at an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √x. Find f(x − 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

2802

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

√(x − 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

T(h) = 20 − 6.5h. Find T(0.3t).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

2803

QUESTION ID

S76-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20 − 1.95t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $f(x) = 1/(x - 1)$, then $f(x^2) = 1/(x^2 - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

2804

QUESTION ID

S76-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q3/5

If $g(x) = x^2 - 9$, then $g(3)$ is

A. -3

B. -6

C. 9

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

2805

QUESTION ID

S76-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $9 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q4/5

2805

ANSWER KEY Reduced copy of the question page, with its answer

p(e) = 1.2e. Find p(45).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

2806

QUESTION ID

S76-B2-Q5

TYPE

Short answer

CORRECT ANSWER

54

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSF-BF.A.1.c

BACKWARD 3

Domains of radical, rational, and log functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{(x - 4)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

2808

QUESTION ID

S76-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log_2 u$ in terms of u.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

2809

QUESTION ID

S76-B3-Q2

TYPE

Short answer

CORRECT ANSWER

u > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q2/5

True or false?

$1/(x^2 - 1)$ is undefined only at $x = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

2810

QUESTION ID

S76-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 1 = 0$ at both $x = 1$ and $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S76-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q3/5

2810

ANSWER KEY Reduced copy of the question page, with its answer

$x^2 - 9 > 0$ when

A. $x < -3$ or $x > 3$

B. $-3 < x < 3$

C. all x

D. $x > 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

2811

QUESTION ID

S76-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $x < -3$ or $x > 3$

GRADING NOTE Outside the roots.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{x} - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

2812

QUESTION ID

S76-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 1

Decompose functions into compositions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

Write $h(x) = (3x + 1)^5$ as $f(g(x))$. Give f and g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

2814

QUESTION ID

S76-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$g(x) = 3x + 1$,
$f(x) = x^5$

GRADING NOTE Any valid decomposition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $h(x) = 2^{x^2}$ as $f(g(x))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

2815

QUESTION ID

S76-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$g(x) = x^2$, $f(x) = 2^x$

GRADING NOTE Any valid decomposition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q2/5

True or false?

Every composition f(g(x)) can be decomposed in only one way.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

2816

QUESTION ID
S76-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Many decompositions exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(x) = x + 1$, $g(x) = 2x$, $k(x) = x^2$, then $f(g(k(3)))$ is

A. 37

B. 19

C. 64

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

2817

QUESTION ID

S76-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 19

GRADING NOTE $k(3) = 9$, $g = 18$, $f = 19$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a context for a composition V(r(t)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

2818

QUESTION ID
S76-F1-Q5

TYPE
Short answer

CORRECT ANSWER
For example the volume of a balloon whose radius r grows with time t

GRADING NOTE Any valid context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSF-BF.A.1.c

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Verify that f(x) = 2x + 3 and g(x) = (x − 3)/2 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

2820

QUESTION ID
S76-F2-Q1

TYPE
Short answer

CORRECT ANSWER
f(g(x)) = x and g(f(x)) = x

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute f(g(x)) for $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

2821

QUESTION ID
S76-F2-Q2

TYPE
Short answer

CORRECT ANSWER
x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q2/5

True or false?

If f(g(x)) = x for all x in g's domain and g(f(x)) = x for all x in f's domain, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

2822

QUESTION ID

S76-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = x + 5 and g(x) = x − 5, f(g(2)) is

A. 2

B. −3

C. 12

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

2823

QUESTION ID

S76-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE Inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why composition is the natural test for inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

2824

QUESTION ID

S76-F2-Q5

TYPE

Short answer

CORRECT ANSWER

An inverse undoes the function, so applying one after the other must return the input.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S76-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 76 • FORWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b • ADVANCED EXTENSION (+)

Verify inverses by composition on the stated domains

MAIN	Algebra II main question	**HSF-BF.B.4.b**
BACKWARD 1	Find inverses algebraically	**HSF-BF.B.4.a**
BACKWARD 2	Compose functions	**HSF-BF.A.1.c**
BACKWARD 3	Square roots of squares	**HSN-RN.A.2**
FORWARD 1	Show that proposed inverses fail	**HSF-BF.B.4.b**
FORWARD 2	Read inverse values from tables and graphs	**HSF-BF.B.4.c**

SET 77 • HSF-BF.B.4.b

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 2x + 3$ and $g(x) = (x - 3)/2$ are inverses?

A. $f(x) \cdot g(x) = (2x + 3)(x - 3)/2 = 1$, so f and g are inverses.

B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

C. $f(1) = 5$ and $g(5) = 1$, so f and g are inverses.

D. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ for all real x, and one composition is enough by itself.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

2827

QUESTION ID
S77-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. $f(g(x)) = 2 \cdot (x - 3)/2 + 3 = x$ and $g(f(x)) = ((2x + 3) - 3)/2 = x$, for all real x.

GRADING NOTE Both compositions must simplify to x. The others check one composition only, one input only, or multiply the functions instead of composing them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = x^3 - 1$ and $g(x) = \sqrt[3]{(x + 1)}$ are inverses?

A. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$, but only for $x \geq 0$, so f and g are inverses only when $x \geq 0$.

B. $g(f(x)) = \sqrt[3]{(x^3 - 1)} + 1 = x$, so f and g are inverses.

C. $f(2) = 7$ and $g(7) = 2$, so f and g are inverses.

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

2828

QUESTION ID

S77-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $f(g(x)) = (\sqrt[3]{(x + 1)})^3 - 1 = x$ and $g(f(x)) = \sqrt[3]{((x^3 - 1) + 1)} = \sqrt[3]{(x^3)} = x$, for all real x.

GRADING NOTE Both compositions simplify to x for every real x (cube roots are defined for negatives). The others add a false domain limit, check one input, or compose incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 2^x$ and $g(x) = \log_2 x$ are inverses, with the correct domains?

A. $f(g(x)) = 2 \cdot \log_2 x = x$ for $x > 0$.

B. $f(g(x)) = 2^{\log_2 x} = x$ and $g(f(x)) = \log_2(2^x) = x$, both for every real number x.

C. $f(3) = 8$ and $g(8) = 3$, so f and g are inverses for all x.

D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

2829

QUESTION ID
S77-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. $f(g(x)) = 2^{\log_2 x} = x$ for $x > 0$, and $g(f(x)) = \log_2(2^x) = x$ for all real x.

GRADING NOTE $\log_2 x$ is defined only for $x > 0$, so $f(g(x)) = x$ holds for $x > 0$; $g(f(x)) = x$ for every real x. The others ignore the domain of $\log_2$, multiply instead of composing, or check one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which work verifies by composition that $f(x) = 1/(x - 2)$ $(x \neq 2)$ and $g(x) = 1/x + 2$ $(x \neq 0)$ are inverses?

A. $f(3) = 1$ and $g(1) = 3$, so f and g are inverses.

B. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$, so f and g are inverses for all real x with no excluded values needed.

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

D. $g(f(x)) = 1/(x - 2) + 2 = (2x - 3)/(x - 2)$, so they are not inverses.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

2830

QUESTION ID

S77-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $f(g(x)) = 1/((1/x + 2) - 2) = 1/(1/x) = x$ for $x \neq 0$, and $g(f(x)) = 1/(1/(x - 2)) + 2 = (x - 2) + 2 = x$ for $x \neq 2$.

GRADING NOTE Both compositions simplify to x on the stated domains. The others drop the second composition and the domains, compute $f(x) + 2$ instead of $g(f(x))$, or check one input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q4/5

$f(x) = x^2$ with domain $x \geq 0$ and $g(x) = \sqrt{x}$. Which work verifies that they are inverses and explains why the domain restriction is needed?

A. The restriction is needed because $\sqrt{x}$ is undefined for $x > 0$.

B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

C. $f(g(x)) = (\sqrt{x})^2 = x$ and $g(f(x)) = \sqrt{(x^2)} = x$ for every real number x, so the two functions are inverses and no domain restriction is needed.

D. $g(f(x)) = \sqrt{(x^2)} = \pm x$, and the restriction chooses the + sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

2831

QUESTION ID
S77-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. $f(g(x)) = (\sqrt{x})^2 = x$ for $x \geq 0$, and $g(f(x)) = \sqrt{(x^2)} = |x| = x$ because $x \geq 0$. Without the restriction, $\sqrt{(x^2)} = |x| \neq x$ for $x < 0$.

GRADING NOTE $\sqrt{(x^2)} = |x|$, which equals x only when $x \geq 0$ (for example $\sqrt{((-3)^2)} = 3$). The others ignore $|x|$, state a false fact about $\sqrt{x}$, or treat $\sqrt{}$ as giving two values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • MAIN • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 1

Find inverses algebraically • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of f(x) = 2x + 3.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

2833

QUESTION ID

S77-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 3)/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = x^3 - 1$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

2834

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$\sqrt[3]{(x + 1)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q2/5

True or false?

The inverse of 2^x is $\log_2 x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

2835

QUESTION ID

S77-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The inverse of f(x) = 1/(x − 2) is

A. 1/x + 2

B. 1/(x + 2)

C. 2 − 1/x

D. x − 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

2836

QUESTION ID

S77-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 1/x + 2

GRADING NOTE Solve y = 1/(x − 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = x^2$ for $x \geq 0$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

2837

QUESTION ID

S77-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt{x}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 2

Compose functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 2x + 3 and g(x) = (x − 3)/2. Compute f(g(5)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

2839

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-B2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 2x + 3 and g(x) = (x − 3)/2. Compute g(f(5)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

2840

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-B2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q2/5

True or false?

f(g(x)) means apply g first, then f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

2841

QUESTION ID

S77-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inside out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^3$ and $g(x) = x + 1$, $f(g(x))$ is

A. $(x + 1)^3$

B. $x^3 + 1$

C. $x + 1$

D. $3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

2842

QUESTION ID

S77-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $(x + 1)^3$

GRADING NOTE Substitute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute $\log_2(2^5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

2843

QUESTION ID
S77-B2-Q5

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSF-BF.B.4.b

BACKWARD 3

Square roots of squares • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\sqrt{((-3)^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

2845

QUESTION ID

S77-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

2846

QUESTION ID

S77-B3-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q3/5

2847

QUESTION ID

S77-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{(x^2)} = |x|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt{(x^2)}$ equals

A. $|x|$

B. x^2

C. $-x$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

2848

QUESTION ID
S77-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $|x|$

GRADING NOTE Principal root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $\sqrt{(x^2)} = x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

2849

QUESTION ID

S77-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-B3-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 1

Show that proposed inverses fail • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Show by composition that f(x) = 3x − 1 and g(x) = (x − 1)/3 are not inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

2851

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = x − 2 ≠ x

GRADING NOTE Form required: a composition that is not x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ (all real x) and $g(x) = \sqrt{x}$. Which composition fails to equal x, and for which x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

2852

QUESTION ID

S77-F1-Q2

TYPE

Short answer

CORRECT ANSWER

g(f(x)) = |x| ≠ x for x < 0

GRADING NOTE Form required: the composition and the x-values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Checking f(g(x)) = x alone is always enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

2853

QUESTION ID
S77-F1-Q3

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Both compositions must be checked (with domains).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = e^x$ and $g(x) = \ln x$, $g(f(x)) =$

A. $|x|$

B. x only for $x > 0$

C. x for all real x

D. $e^{\ln x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

2854

QUESTION ID
S77-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. x for all real x

GRADING NOTE $\ln(e^x) = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Fix the domain of $f(x) = (x - 1)^2$ so that $g(x) = 1 + \sqrt{x}$ is its inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

2855

QUESTION ID

S77-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSF-BF.B.4.b

FORWARD 2

Read inverse values from tables and graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(7)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

2857

QUESTION ID
S77-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $f^{-1}(f^{-1}(15))$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

2858

QUESTION ID
S77-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(3) = 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q2/5

True or false?

If (a, b) is on the graph of f, then (b, a) is on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

2859

QUESTION ID

S77-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(4) = 9$, then $f^{-1}(9)$ is

A. 1/4

B. 4

C. 13

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

2860

QUESTION ID

S77-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(1) = 3 and f(2) = 7. Explain how these values give $f(f^{-1}(7)) = 7$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

2861

QUESTION ID
S77-F2-Q5

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(7) = 2$ and $f(2) = 7$, so $f(f^{-1}(7)) = 7$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S77-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 77 • FORWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c • ADVANCED EXTENSION (+)

Read values of an inverse function from graphs and tables

MAIN	Algebra II main question	**HSF-BF.B.4.c**
BACKWARD 1	Inverse functions swap inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 2	Read points from graphs and tables	**8.F.A.1**
BACKWARD 3	One-to-one functions	**HSF-BF.B.4.b**
FORWARD 1	Estimate inverse values between data points and sketch inverse...	**HSF-BF.B.4.c**
FORWARD 2	Restrict the domain to make a function invertible	**HSF-BF.B.4.d**

SET 78 • HSF-BF.B.4.c

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

The table shows f. Find $f^{-1}(15)$ and explain.

x	0	1	2	3	4
f(x)	1	3	7	15	31

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

2864

QUESTION ID

S78-M-Q1

TYPE

Short answer

CORRECT ANSWER

3: f(3) = 15

GRADING NOTE Form required: value and reason.

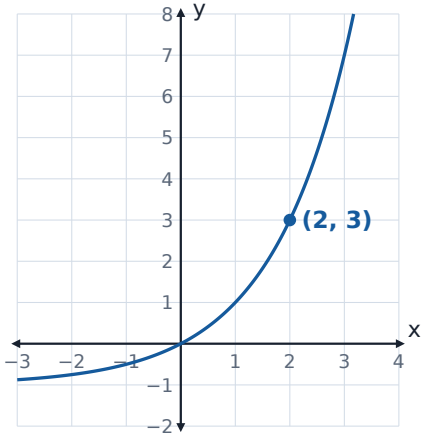
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = 2^x - 1$ is shown. Use it to find $f^{-1}(3)$.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

2865

QUESTION ID
S78-M-Q2

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. The point (2, 3) is on the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a decreasing function g. Find $g^{-1}(10)$ and $g^{-1}(g(2))$.

x	1	2	3	4	5
g(x)	40	25	16	10	4

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

2866

QUESTION ID
S78-M-Q3

TYPE
Short answer

CORRECT ANSWER
$g^{-1}(10) = 4$;
$g^{-1}(g(2)) = 2$

GRADING NOTE Form required: both.

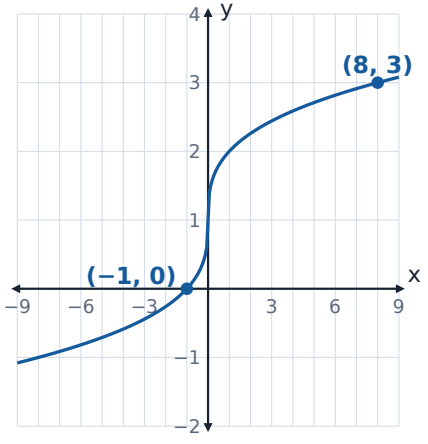
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $f(x) = \sqrt[3]{x} + 1$ is shown. Use it to find $f^{-1}(0)$ and $f^{-1}(3)$.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

2867

QUESTION ID

S78-M-Q4

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(0) = -1$;

$f^{-1}(3) = 8$

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The table gives a plane's altitude A (m) t minutes after takeoff. Find $A^{-1}(1200)$ and interpret it.

t (min)	0	1	2	3	4	5
A (m)	0	250	550	880	1200	1500

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

2868

QUESTION ID

S78-M-Q5

TYPE

Short answer

CORRECT ANSWER

4 minutes: the plane reaches 1200 m four minutes after takeoff

GRADING NOTE Form required: value and interpretation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • MAIN • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 1

Inverse functions swap inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

If $f(3) = 15$, what is $f^{-1}(15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

2870

QUESTION ID

S78-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $f(x) = 2^x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

2871

QUESTION ID

S78-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = \log_2(x + 1)$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$f^{-1}(y)$ is the input of f that gives the output y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

2872

QUESTION ID

S78-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If (5, 4) is on the graph of g, which point is on g^{-1}?

A. (5, 4)

B. (1/5, 1/4)

C. (−5, −4)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

2873

QUESTION ID

S78-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 5)

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use the formula to find $f^{-1}(3)$ for $f(x) = 2^x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

2874

QUESTION ID

S78-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 78 • BACKWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 2

Read points from graphs and tables • Grade 8 • Functions • 8.F.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. What is f(2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

2876

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plane's altitude A (m) at t = 0, 1, 2, 3, 4 minutes is 0, 250, 550, 880, 1200. What is A(3)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

2877

QUESTION ID

S78-B2-Q2

TYPE

Short answer

CORRECT ANSWER

880 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q2/5

True or false?

A function gives exactly one output for each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

2878

QUESTION ID

S78-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Which input of g gives 16?

A. 16

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

2879

QUESTION ID

S78-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

GRADING NOTE Read backward.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph passes through (8, 3). What is the output at input 8?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

2880

QUESTION ID

S78-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 78 • BACKWARD 2 • Q5/5

SET 78 • HSF-BF.B.4.c

BACKWARD 3

One-to-one functions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Why does a strictly increasing function have an inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

2882

GRADING NOTE Any equivalent answer.

QUESTION ID

S78-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Different inputs give different outputs, so each output comes from one input

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Does a function with f(1) = 4 and f(3) = 4 have an inverse function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

2883

QUESTION ID
S78-B3-Q2

TYPE
Short answer

CORRECT ANSWER
No: 4 has two inputs

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-B3-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q2/5

True or false?

A decreasing function can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

2884

QUESTION ID

S78-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Still one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which table has an inverse function?

A. x: 1, 2, 3; y: 5, 7, 5

B. x: 1, 2, 3; y: 5, 7, 9

C. x: 1, 2, 3; y: 0, 1, 0

D. x: 1, 2, 3; y: 2, 2, 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

2885

QUESTION ID
S78-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. x: 1, 2, 3; y: 5, 7, 9

GRADING NOTE Distinct outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(2) = 7 and f(3) = 15. Verify $f(f^{-1}(15)) = 15$ with the table.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

2886

QUESTION ID

S78-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(15) = 3$ and $f(3) = 15$

GRADING NOTE Form required: the check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 78 • BACKWARD 3 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 1

Estimate inverse values between data points and sketch inverse graphs • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

A plane's altitude is 880 m at t = 3 min and 1200 m at t = 4 min. Estimate the time when it reaches 1000 m.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

2888

QUESTION ID

S78-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 3.4 minutes

GRADING NOTE Accept 3.3 to 3.5 (linear interpolation 3 + 120/320).

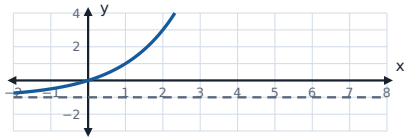
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q1/5

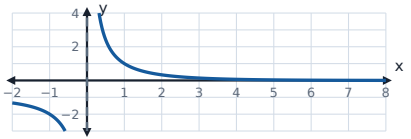
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows the inverse of $f(x) = 2^x - 1$?

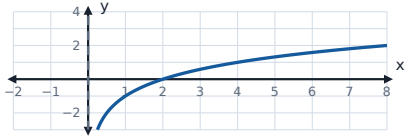
A



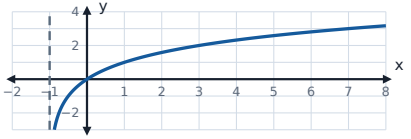
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

2889

QUESTION ID

S78-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Swap the coordinates of points of f: (0, 0), (1, 1), (2, 3) on f give (0, 0), (1, 1), (3, 2) on f^{-1}; the asymptote $y = -1$ becomes $x = -1$; $f^{-1}(x) = \log_2(x + 1)$. The others show f itself, the reciprocal $1/f(x)$, and $\log_2 x - 1$ (operations undone in the wrong order).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q2/5

True or false?

The graph of f^{-1} is the reflection of the graph of f across y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

2890

QUESTION ID

S78-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Swap coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A decreasing function has $g(1) = 40$, $g(2) = 25$, $g(3) = 16$, $g(4) = 10$. Estimate $g^{-1}(20)$.

A. between 1 and 2

B. between 2 and 3

C. between 4 and 5

D. between 3 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

2891

QUESTION ID

S78-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. between 2 and 3

GRADING NOTE $g(2) = 25$, $g(3) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(0) = 1, f(1) = 3, f(2) = 7, f(3) = 15. Find $(f^{-1})^{-1}(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

2892

QUESTION ID
S78-F1-Q5

TYPE
Short answer

CORRECT ANSWER
f(2) = 7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.c • Set 78 • FORWARD 1 • Q5/5

SET 78 • HSF-BF.B.4.c

FORWARD 2

Restrict the domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

$f(x) = (x - 3)^2$ is not invertible. Give a domain restriction that makes it invertible, and write the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

2894

QUESTION ID

S78-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 3$; $f^{-1}(x) = 3 + \sqrt{x}$

GRADING NOTE Any valid restriction with its inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S78-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table of h shows $h(-2) = 4$, $h(0) = 0$, $h(2) = 4$. Explain why $h^{-1}(4)$ is not defined and how to fix it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

2895

QUESTION ID

S78-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 has two inputs; restrict h to x ≥ 0 (or x ≤ 0)

GRADING NOTE Form required: reason and fix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q2/5

True or false?

Restricting the domain can turn a non-invertible function into an invertible one.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

2896

QUESTION ID
S78-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE One-to-one on the piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ restricted to $x \le 0$ has inverse

A. $\sqrt{x}$

B. x^2

C. $-\sqrt{x}$

D. $1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

2897

QUESTION ID

S78-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-\sqrt{x}$

GRADING NOTE Negative outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q4/5

Why must the restricted piece still take every output value of the original function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

2898

QUESTION ID

S78-F2-Q5

TYPE

Short answer

CORRECT ANSWER

To keep the same range, the restricted function should reach all the outputs (for example x ≥ 3 still gives all y ≥ 0).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S78-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 78 • FORWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d • ADVANCED EXTENSION (+)

Restrict a domain to make a function invertible and write the inverse

MAIN	Algebra II main question	**HSF-BF.B.4.d**
BACKWARD 1	Find the vertex of a quadratic	**HSF-IF.C.8.a**
BACKWARD 2	Write inverses of one-to-one functions	**HSF-BF.B.4.a**
BACKWARD 3	Domains and ranges	**HSF-IF.B.5**
FORWARD 1	Compare different restrictions	**HSF-BF.B.4.d**
FORWARD 2	Restrict trigonometric functions to define inverses	**HSF-TF.B.6**

SET 79 • HSF-BF.B.4.d

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x - 3)^2$ to $x \geq 3$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

2901

QUESTION ID

S79-M-Q1

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = \sqrt{x} + 3,\ x \geq 0$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = x^2 + 1$ to $x \le 0$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

2902

QUESTION ID

S79-M-Q2

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = -\sqrt{(x - 1)}$, $x \ge 1$

GRADING NOTE Form required: inverse (with the negative root) and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = |x + 2|$ to $x \geq -2$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

2903

QUESTION ID

S79-M-Q3

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = x - 2, x \geq 0$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q3**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x + 1)^2 - 4$ to $x \geq -1$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

2904

QUESTION ID
S79-M-Q4

TYPE
Short answer

CORRECT ANSWER
$f^{-1}(x) = -1 + \sqrt{(x + 4)}$, $x \geq -4$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = x^4$ to $x \geq 0$. Write f^{-1} and its domain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

2905

QUESTION ID

S79-M-Q5

TYPE

Short answer

CORRECT ANSWER

$f^{-1}(x) = \sqrt[4]{x},\ x \geq 0$

GRADING NOTE Form required: inverse and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • MAIN • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 1

Find the vertex of a quadratic • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertex of $y = (x + 1)^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

2907

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, −4)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertex of $y = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

2908

QUESTION ID

S79-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q2/5

True or false?

A parabola is increasing on one side of its vertex and decreasing on the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

2909

QUESTION ID

S79-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Turning point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The vertex of $y = (x - 3)^2$ is

A. (0, 9)

B. (0, 3)

C. (−3, 0)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

2910

QUESTION ID

S79-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (3, 0)

GRADING NOTE h = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

On which side of the vertex is $y = (x - 3)^2$ increasing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

2911

QUESTION ID

S79-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$x \geq 3$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 79 • BACKWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 2

Write inverses of one-to-one functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = (x - 3)^2$ for x when $x \geq 3$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

2913

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$3 + \sqrt{y}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = x^2 + 1$ for x when $x \leq 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

2914

QUESTION ID

S79-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-\sqrt{(y - 1)}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q2/5

True or false?

When $x \leq 0$, the solution of $x^2 = 4$ is $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

2915

QUESTION ID

S79-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Choose the negative root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving $y = x + 2$ $(x \geq -2)$ for x gives

A. $2 - y$

B. $y + 2$

C. $y - 2$

D. $-y - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

2916

QUESTION ID

S79-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $y - 2$

GRADING NOTE Linear piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = x^4$ for x when $x \geq 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

2917

QUESTION ID

S79-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt[4]{y}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 79 • BACKWARD 2 • Q5/5

SET 79 • HSF-BF.B.4.d

BACKWARD 3

Domains and ranges • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = (x - 3)^2$ for $x \geq 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

2919

QUESTION ID

S79-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = (x + 1)^2 - 4$ for $x \geq -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

2920

QUESTION ID

S79-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$y \geq -4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The domain of the inverse is the range of the restricted function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

2921

QUESTION ID

S79-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q3/5

2921

ANSWER KEY Reduced copy of the question page, with its answer

The range of $x^2 + 1$ for $x \leq 0$ is

A. $y \geq 0$

B. $y \geq 1$

C. all reals

D. $y \leq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

2922

QUESTION ID

S79-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. y ≥ 1

GRADING NOTE Minimum 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $|x + 2|$ for $x \geq -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

2923

QUESTION ID

S79-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$y \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 79 • BACKWARD 3 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 1

Compare different restrictions • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Restrict $f(x) = (x - 3)^2$ to $x \leq 3$. Write the inverse.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

2925

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3 - \sqrt{x}, x \geq 0$

GRADING NOTE Form required: formula and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -5(t - 2)^2 + 20$ for $0 \le t \le 4$. Which restriction gives the time as a function of height on the way down? Write it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

2926

QUESTION ID
S79-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$2 \le t \le 4$;
$t = 2 + \sqrt{((20 - h)/5)}$

GRADING NOTE Form required: restriction and inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q2/5

True or false?

A function can have more than one valid restriction that makes it invertible.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

2927

QUESTION ID
S79-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Either side of the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which restriction does not make $f(x) = x^2$ invertible?

A. $x \geq 0$

B. $x \leq 0$

C. $-1 \leq x \leq 1$

D. $1 \leq x \leq 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

2928

QUESTION ID

S79-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $-1 \leq x \leq 1$

GRADING NOTE Contains both ±1/2, for example.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why is the restriction $x \geq 0$ natural for $f(x) = x^2$ in the area formula $A = s^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

2929

QUESTION ID

S79-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Side lengths are nonnegative.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 79 • FORWARD 1 • Q5/5

SET 79 • HSF-BF.B.4.d

FORWARD 2

Restrict trigonometric functions to define inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Why is sin x not invertible on all real numbers?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

2931

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

It repeats values (for example sin 0 = sin π = 0), so it is not one-to-one.

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the standard restricted domain of sin x and explain why it works.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

2932

QUESTION ID

S79-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−π/2 ≤ x ≤ π/2: sin is increasing there and takes every value from −1 to 1 once.

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q2/5

True or false?

cos x is restricted to $0 \leq x \leq \pi$ to define arccos.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

2933

QUESTION ID

S79-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Decreasing there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arcsin(1) equals

A. 0

B. 3π/2

C. π/2

D. π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

2934

QUESTION ID

S79-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π/2

GRADING NOTE In the restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S79-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $0 \le x \le \pi$ would not work for sine.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

2935

GRADING NOTE Any equivalent explanation.

QUESTION ID

S79-F2-Q5

TYPE

Short answer

CORRECT ANSWER

sin increases then decreases on [0, π], so values repeat (sin(π/6) = sin(5π/6)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S79-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 79 • FORWARD 2 • Q5/5

SET 80 • HSF-BF.B.5 • ADVANCED EXTENSION (+)

Use the inverse relationship of exponents and logarithms to solve equations

MAIN	Algebra II main question	**HSF-BF.B.5**
BACKWARD 1	Write exact logarithmic solutions of exponential equations	**HSF-LE.A.4**
BACKWARD 2	Check for extraneous solutions	**HSA-REI.A.2**
BACKWARD 3	Domains of logarithmic expressions	**HSF-IF.B.5**
FORWARD 1	Use log properties and change of base in equations	**HSF-BF.B.5**
FORWARD 2	Check logarithmic solutions graphically	**HSA-REI.D.11**

SET 80 • HSF-BF.B.5

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_3 x = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

2938

QUESTION ID

S80-M-Q1

TYPE

Short answer

CORRECT ANSWER

81

GRADING NOTE Any equivalent form. $x = 3^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\log_5(x - 1) = 4$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

2939

QUESTION ID

S80-M-Q2

TYPE

Short answer

CORRECT ANSWER

26

GRADING NOTE Any equivalent form. $\log_5(x - 1) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ln(x + 2) = 1 exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

2940

QUESTION ID

S80-M-Q3

TYPE

Short answer

CORRECT ANSWER

e − 2

GRADING NOTE Form required: exact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2 x + \log_2(x - 2) = 3$ and check for extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

2941

QUESTION ID
S80-M-Q4

TYPE
Short answer

CORRECT ANSWER
x = 4; x = −2 is extraneous

GRADING NOTE Form required: solution and rejection. $x(x - 2) = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve log x − log(x − 9) = 1 (base 10) and check.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

2942

QUESTION ID

S80-M-Q5

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Form required: the solution with a check. x/(x − 9) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 1

Write exact logarithmic solutions of exponential equations • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3^x = 81$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

2944

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $e^x = 5$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

2945

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

ln 5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-B1-Q2**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log_b y = x$ means $b^x = y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

2946

QUESTION ID

S80-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$5^2 = 25$ written as a logarithm is

A. $\log_5 2 = 25$

B. $\log_2 25 = 5$

C. $\log_{25} 5 = 2$

D. $\log_5 25 = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

2947

QUESTION ID

S80-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $\log_5 25 = 2$

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $10^x = 1000$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

2948

QUESTION ID

S80-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 2

Check for extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = -2$ in $\log_2 x + \log_2(x - 2) = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

2950

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$\log_2(-2)$ is undefined, so -2 is extraneous

GRADING NOTE Form required: check and conclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Check $x = 4$ in $\log_2 x + \log_2(x - 2) = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

2951

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2 + 1 = 3; a solution

GRADING NOTE Form required: check.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q2/5

True or false?

Combining logs can introduce solutions outside the original domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

2952

QUESTION ID

S80-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE log a + log b = log(ab) needs a, b > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is excluded from log(x − 9)?

A. x = 20

B. x = 100

C. x = 5

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

2953

QUESTION ID

S80-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x = 5

GRADING NOTE x − 9 > 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x(x - 2) = 8$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

2954

QUESTION ID

S80-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4 or −2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.B.5

BACKWARD 3

Domains of logarithmic expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log_5(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

2956

QUESTION ID

S80-B3-Q1

TYPE

Short answer

CORRECT ANSWER

x > 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of ln(x + 2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

2957

QUESTION ID

S80-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x > −2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

log 0 = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

2958

QUESTION ID

S80-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No power of 10 equals 0, so log 0 is undefined (log 1 = 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log_2 x + \log_2(x - 2)$ is

A. $x > 2$

B. $x \neq 2$

C. $x > 0$

D. all reals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

2959

QUESTION ID

S80-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x > 2

GRADING NOTE Both inputs positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q4/5

2959

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\log x - \log(x - 9)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

2960

QUESTION ID

S80-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$x > 9$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 80 • BACKWARD 3 • Q5/5

2960

SET 80 • HSF-BF.B.5

FORWARD 1

Use log properties and change of base in equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2(x + 6) - \log_2 x = 2$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

2962

QUESTION ID

S80-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. $(x + 6)/x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_4 x = \log_2 3$. (Hint: change of base.)

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

2963

QUESTION ID

S80-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. $\log_4 x = (\log_2 x)/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q2/5

2963

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log_2(x + 6) - \log_2 x = \log_2((x + 6)/x)$ for $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q3/5

2964

QUESTION ID

S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Quotient property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\log(x + 1) + \log(x - 1) = \log 8$ has solution

A. no solution

B. $x = 8$

C. $x = 3$

D. $x = -3$ and 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

2965

QUESTION ID

S80-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x = 3$

GRADING NOTE $x^2 - 1 = 8$; -3 rejected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\log_2(x - 5) = \log_2(1 - x)$ has no solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

2966

QUESTION ID

S80-F1-Q5

TYPE

Short answer

CORRECT ANSWER

It needs x > 5 and x < 1 at the same time, which is impossible.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.B.5

FORWARD 2

Check logarithmic solutions graphically • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the graphs of $y = \log_2 x + \log_2(x - 2)$ and $y = 3$ show the solution $x = 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

2968

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

They intersect only at x = 4; the left graph exists only for x > 2.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graph or table to approximate the solution of ln $x = 2 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

2969

QUESTION ID

S80-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.56

GRADING NOTE Accept 1.55 to 1.56 (root 1.5571).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q2/5

True or false?

An extraneous solution of a log equation corresponds to no intersection point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

2970

QUESTION ID

S80-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outside the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of y = log x and y = 1 meet at

A. x = e

B. x = 10

C. x = 1

D. x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

2971

QUESTION ID

S80-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = 10

GRADING NOTE log 10 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can ln x = 2 − x not be solved exactly with logarithm rules?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

2972

QUESTION ID
S80-F2-Q5

TYPE
Short answer

CORRECT ANSWER
x appears both inside a logarithm and outside it; no algebraic step isolates x.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S80-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 80 • FORWARD 2 • Q5/5

SET 81 • HSF-TF.A.3 • ADVANCED EXTENSION (+)

Derive values from special triangles and relate π − x, π + x, 2π − x to x

MAIN	Algebra II main question	**HSF-TF.A.3**
BACKWARD 1	Special right triangle side ratios	**HSG-SRT.C.6**
BACKWARD 2	Unit-circle values in all quadrants	**HSF-TF.A.2**
BACKWARD 3	The Pythagorean Theorem	**8.G.B.7**
FORWARD 1	Derive more special values and the reflection identities	**HSF-TF.A.3**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 81 • HSF-TF.A.3

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

A 30°−60°−90° triangle with hypotenuse 1 is half of an equilateral triangle with side 1. Which derivation of sin(π/3) and cos(π/3) is correct?

A. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the long leg, so sin(π/3) = √3/2 and cos(π/3) = 1/2.

B. The long leg is √(1 + 1/4) = √5/2, so sin(π/3) = √5/2.

C. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the short leg, so sin(π/3) = 1/2 and cos(π/3) = √3/2.

D. The legs are 1/2 and 1 − 1/2 = 1/2, so sin(π/3) = cos(π/3) = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

2975

QUESTION ID

S81-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. The short leg is 1/2 and the long leg is √(1 − 1/4) = √3/2. The angle π/3 is opposite the long leg, so sin(π/3) = √3/2 and cos(π/3) = 1/2.

GRADING NOTE Halving the base gives 1/2; the Pythagorean Theorem gives √3/2; the larger angle is opposite the longer leg. The others put π/3 opposite the short leg, subtract lengths instead of using the Pythagorean Theorem, or add instead of subtracting squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which derivation of tan(π/4) from a 45°−45°−90° triangle is correct?

A. The hypotenuse is twice a leg, so tan(π/4) = 1/2.

B. tan(π/4) = sin(π/4) + cos(π/4) = √2.

C. The legs are 1 and √2, so tan(π/4) = opposite/adjacent = 1/√2 = √2/2.

D. The two legs are equal, so tan(π/4) = opposite/adjacent = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

2976

QUESTION ID

S81-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. The two legs are equal, so tan(π/4) = opposite/adjacent = 1.

GRADING NOTE Equal legs make the ratio 1. The others use the hypotenuse as a leg, add instead of dividing, or use a 30°−60°−90° fact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use sin(π − x) = sin x to find sin(5π/6) exactly, and explain the identity with the unit circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

2977

QUESTION ID

S81-M-Q3

TYPE

Short answer

CORRECT ANSWER

sin(5π/6) = sin(π/6) = 1/2: the points at x and π − x are reflections over the y-axis, with the same y-coordinate

GRADING NOTE Form required: value and explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use cos(π + x) = −cos x to find cos(7π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

2978

QUESTION ID

S81-M-Q4

TYPE

Short answer

CORRECT ANSWER

−cos(π/6) = −√3/2

GRADING NOTE Form required: exact value and the relation used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use tan(2π − x) = −tan x to find tan(5π/3) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

2979

QUESTION ID

S81-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-\tan(\pi/3) = -\sqrt{3}$

GRADING NOTE Form required: exact value and the relation used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • MAIN • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 1

Special right triangle side ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Give the sides of a 30°−60°−90° triangle with hypotenuse 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

2981

QUESTION ID
S81-B1-Q1

TYPE
Short answer

CORRECT ANSWER
1, √3, 2

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the hypotenuse of a 45°−45°−90° triangle with legs 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

2982

QUESTION ID
S81-B1-Q2

TYPE
Short answer

CORRECT ANSWER
√2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q2/5

True or false?

In a 30°−60°−90° triangle, the shortest side is half the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

2983

QUESTION ID

S81-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Half of an equilateral triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S81-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos 30° equals

A. √3

B. √2/2

C. 1/2

D. √3/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

2984

QUESTION ID
S81-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. √3/2

GRADING NOTE Adjacent/hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give tan 60°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

2985

QUESTION ID

S81-B1-Q5

TYPE

Short answer

CORRECT ANSWER

√3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B1-Q5**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 2

Unit-circle values in all quadrants • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(5π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

2987

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−√3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B2-Q1**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(7π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

2988

QUESTION ID

S81-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

sin is positive in quadrant II.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

2989

QUESTION ID

S81-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $y > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q3/5

2989

ANSWER KEY Reduced copy of the question page, with its answer

Which quadrant contains 5π/3?

A. II

B. I

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

2990

QUESTION ID

S81-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. IV

GRADING NOTE Between 3π/2 and 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give tan(π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

2991

QUESTION ID

S81-B2-Q5

TYPE

Short answer

CORRECT ANSWER

√3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-B2-Q5**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-TF.A.3

BACKWARD 3

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A right triangle has hypotenuse 1 and leg 1/2. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

2993

QUESTION ID

S81-B3-Q1

TYPE

Short answer

CORRECT ANSWER

√3/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A right isosceles triangle has hypotenuse 1. Find each leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

2994

QUESTION ID

S81-B3-Q2

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(1/2)^2 + (\sqrt{3}/2)^2 = 2.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

2995

QUESTION ID

S81-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1/4 + 3/4 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Legs 1 and 1 give hypotenuse

A. 2

B. √3

C. √2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

2996

QUESTION ID

S81-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. √2

GRADING NOTE √2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find x if $x^2 + x^2 = 1$ ($x > 0$).

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

2997

QUESTION ID

S81-B3-Q5

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 1

Derive more special values and the reflection identities • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Which derivation of tan(π/6) from a special triangle is correct?

A. The opposite side is half the hypotenuse, so tan(π/6) = 1/2.

B. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg 1/2 and adjacent to the leg √3/2, so tan(π/6) = (1/2)/(√3/2) = 1/√3 = √3/3.

C. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg √3/2 and adjacent to the leg 1/2, so tan(π/6) = (√3/2)/(1/2) = √3 exactly.

D. tan(π/6) = (1/2)/(√3/2) = √3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

2999

QUESTION ID

S81-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. In the 30°−60°−90° triangle with hypotenuse 1, π/6 is opposite the leg 1/2 and adjacent to the leg √3/2, so tan(π/6) = (1/2)/(√3/2) = 1/√3 = √3/3.

GRADING NOTE Opposite over adjacent, then simplify (1/2)/(√3/2) = 1/√3 = √3/3. The others use the angle π/3, give sin(π/6) instead, or simplify the quotient incorrectly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle argument explains why $\cos(\pi - x) = -\cos x$?

A. The points for x and $\pi - x$ are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

B. $\cos(\pi - \pi/3) = -1/2 = -\cos(\pi/3)$, so the rule always holds.

C. $\cos(\pi - x) = \cos \pi - \cos x = -1 - \cos x$.

D. The points for x and $\pi - x$ are reflections of each other over the x-axis, so their x-coordinates (the cosines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

3000

QUESTION ID

S81-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. The points for x and π − x are reflections of each other over the y-axis, so their x-coordinates (the cosines) are opposites.

GRADING NOTE Reflection over the y-axis changes the sign of x and keeps y. The others name the wrong reflection (it keeps x), distribute cosine over a difference, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(\pi + x) = -\sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

3001

QUESTION ID

S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotation by π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos(2π − x) equals

A. cos x

B. −cos x

C. sin x

D. −sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

3002

QUESTION ID

S81-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos x

GRADING NOTE Reflection over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(11π/6) using one of the relations.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

3003

QUESTION ID

S81-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−sin(π/6) = −1/2

GRADING NOTE Form required: value and relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.3 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-TF.A.3

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Explain why sin(−x) = −sin x with the unit circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

3005

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x and −x give points reflected over the x-axis; their y-coordinates are opposites.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is cos an even or odd function? Justify.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

3006

QUESTION ID

S81-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Even: cos(−x) = cos x (same x-coordinate)

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(x + 2\pi) = \sin x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

3007

QUESTION ID

S81-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

tan(−π/4) equals

A. undefined

B. −1

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

3008

QUESTION ID

S81-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. −1

GRADING NOTE Odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $2\pi - x = -x + 2\pi$ to explain why $\tan(2\pi - x) = -\tan x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

3009

QUESTION ID

S81-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$\tan(-x + 2\pi) = \tan(-x)$ by periodicity, and tan is odd, so it equals $-\tan x$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S81-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 81 • FORWARD 2 • Q5/5

SET 82 • HSF-TF.A.4 • ADVANCED EXTENSION (+)

Explain the symmetry and periodicity of trigonometric functions with the unit circle

MAIN	Algebra II main question	**HSF-TF.A.4**
BACKWARD 1	Explain the unit-circle definition	**HSF-TF.A.2**
BACKWARD 2	Even and odd functions	**HSF-BF.B.3**
BACKWARD 3	Reflect points over the axes	**8.G.A.3**
FORWARD 1	Use periodicity and symmetry to evaluate and simplify	**HSF-TF.A.4**
FORWARD 2	Restrict trigonometric functions so they have inverses	**HSF-TF.B.6**

SET 82 • HSF-TF.A.4

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why $\sin(-x) = -\sin x$?

A. The points for x and −x are (a, b) and (−a, b), reflections of each other over the y-axis, so their sines are opposites.

B. $\sin(-\pi/6) = -1/2 = -\sin(\pi/6)$, so the rule always holds.

C. The points for x and −x are (a, b) and (a, −b), reflections over the x-axis, so the sines b and −b are opposites.

D. A negative sign can always be moved outside a function, so $\sin(-x) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

3012

QUESTION ID

S82-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The points for x and −x are (a, b) and (a, −b), reflections over the x-axis, so the sines b and −b are opposites.

GRADING NOTE Equal arcs in opposite directions from (1, 0) give points that mirror over the x-axis. The others name the wrong reflection, use a false general rule ($\cos(-x) = \cos x$, not $-\cos x$), or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why cos(x + 2π) = cos x?

A. Adding 2π moves the point to the opposite side of the unit circle, and opposite points on the circle have the same x-coordinate, so the cosine is the same.

B. cos(x + 2π) = cos x + cos 2π = cos x + 1.

C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

D. cos(0 + 2π) = 1 = cos 0, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

3013

QUESTION ID

S82-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Adding 2π travels one full turn around the unit circle and ends at the same point, so the x-coordinate (the cosine) is the same.

GRADING NOTE The circumference is 2π, so x and x + 2π give the same point. The others distribute cosine over a sum, describe the wrong motion, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which unit-circle explanation shows why $\tan(x + \pi) = \tan x$?

A. Adding π gives the point (a, −b), a reflection over the x-axis, so $\tan(x + \pi) = -b/a = -\tan x$, which is not tan x.

B. Adding π gives the opposite point (−a, −b) of (a, b), and $\tan = (-b)/(-a) = b/a$, the same ratio.

C. Adding π travels a full turn, so the point is the same.

D. $\tan(\pi/4 + \pi) = 1 = \tan(\pi/4)$, so the rule always holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

3014

QUESTION ID
S82-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. Adding π gives the opposite point (−a, −b) of (a, b), and tan = (−b)/(−a) = b/a, the same ratio.

GRADING NOTE Half a turn takes (a, b) to (−a, −b); both coordinates change sign, so the ratio is unchanged. The others use the wrong point, mistake π for a full turn, or check one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Use symmetry to evaluate cos(−2π/3), and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

3015

QUESTION ID

S82-M-Q4

TYPE

Short answer

CORRECT ANSWER

−1/2: cos is even, so cos(−2π/3) = cos(2π/3) = −1/2

GRADING NOTE Form required: value and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is f(x) = sin x cos x even, odd, or neither? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

3016

QUESTION ID

S82-M-Q5

TYPE

Short answer

CORRECT ANSWER

Odd: f(−x) = (−sin x)(cos x) = −f(x)

GRADING NOTE Form required: decision and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • MAIN • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 1

Explain the unit-circle definition • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

What point represents angle x on the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

3018

QUESTION ID

S82-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(cos x, sin x)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which direction does a negative angle go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

3019

QUESTION ID

S82-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Clockwise

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

tan x = sin x/cos x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

3020

QUESTION ID

S82-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q3/5

3020

The point for angle π is

A. (0, −1)

B. (−1, 0)

C. (0, 1)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

3021

QUESTION ID

S82-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 0)

GRADING NOTE Half turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

3022

QUESTION ID

S82-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 2

Even and odd functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Define an even function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

3024

QUESTION ID

S82-B2-Q1

TYPE

Short answer

CORRECT ANSWER

f(−x) = f(x) for every x in the domain

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Define an odd function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

3025

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

f(−x) = −f(x) for every x in the domain

GRADING NOTE Any equivalent definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

An odd function's graph is symmetric about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

3026

QUESTION ID

S82-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Rotation by 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

x^3 is

A. odd

B. neither

C. even

D. both

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q4/5

3027

QUESTION ID

S82-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. odd

GRADING NOTE $(-x)^3 = -x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is the product of an odd and an even function odd or even?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

3028

QUESTION ID

S82-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Odd

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-TF.A.4

BACKWARD 3

Reflect points over the axes • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Reflect (a, b) over the x-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

3030

QUESTION ID

S82-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(a, −b)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q1/5

3030

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (a, b) by 180° about the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

3031

QUESTION ID

S82-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(−a, −b)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q2/5

True or false?

Reflecting over the x-axis keeps the y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

3032

QUESTION ID

S82-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE (a, b) → (a, −b): the x-coordinate stays and y changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q3/5

Reflecting (0.6, 0.8) over the x-axis gives

A. (0.8, 0.6)

B. (−0.6, 0.8)

C. (−0.6, −0.8)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

3033

QUESTION ID

S82-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (0.6, −0.8)

GRADING NOTE Negate y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (√3/2, 1/2) by 180°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

3034

QUESTION ID

S82-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(−√3/2, −1/2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 82 • BACKWARD 3 • Q5/5

3034

SET 82 • HSF-TF.A.4

FORWARD 1

Use periodicity and symmetry to evaluate and simplify • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(17π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

3036

QUESTION ID

S82-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE Any equivalent form. 17π/6 − 2π = 5π/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find tan(−11π/4) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

3037

QUESTION ID

S82-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. tan(−11π/4) = tan(−11π/4 + 3π) = tan(π/4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos(x + \pi) = -\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

3038

QUESTION ID

S82-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Opposite point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sin(x + \pi) + \sin x$ equals

A. 1

B. $\sin 2x$

C. $2 \sin x$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

3039

QUESTION ID

S82-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $\sin(x + \pi) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify cos(−x) + cos(x + 2π).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

3040

QUESTION ID

S82-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2cos x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSF-TF.A.4

FORWARD 2

Restrict trigonometric functions so they have inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Explain how periodicity prevents sin x from having an inverse on all real numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

3042

QUESTION ID

S82-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each value repeats every 2π, so outputs do not determine inputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the restricted domain used for tan x and explain it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

3043

QUESTION ID

S82-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−π/2 < x < π/2: tan increases there and takes every real value once.

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q2/5

3043

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

arccos gives values between 0 and π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

3044

QUESTION ID
S82-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Restricted range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q3/5

arctan(−1) equals

A. 7π/4

B. 3π/4

C. −π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

3045

QUESTION ID

S82-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −π/4

GRADING NOTE In (−π/2, π/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S82-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use oddness to explain why arcsin(−1/2) = −arcsin(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

3046

QUESTION ID

S82-F2-Q5

TYPE

Short answer

CORRECT ANSWER

sin is odd on [−π/2, π/2], so sin(−π/6) = −1/2 and arcsin(−1/2) = −π/6 = −arcsin(1/2).

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S82-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 82 • FORWARD 2 • Q5/5

SET 83 • HSF-TF.B.6 • ADVANCED EXTENSION (+)

Choose and justify domain restrictions that make trigonometric functions invertible

MAIN	Algebra II main question	**HSF-TF.B.6**
BACKWARD 1	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**
BACKWARD 2	Exact values on the unit circle	**HSF-TF.A.2**
BACKWARD 3	Periodicity of trigonometric functions	**HSF-TF.A.4**
FORWARD 1	Compose trigonometric functions with their inverses	**HSF-TF.B.6**
FORWARD 2	Solve trigonometric equations with inverses	**HSF-TF.B.7**

SET 83 • HSF-TF.B.6

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

Explain why sin x has no inverse on all real numbers, and give a restricted domain on which it does.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

3049

QUESTION ID

S83-M-Q1

TYPE

Short answer

CORRECT ANSWER

sin repeats values (not one-to-one); on $-\pi/2 \le x \le \pi/2$ it is increasing and takes every value in $[-1, 1]$ once

GRADING NOTE Form required: reason and domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the standard restricted domain of cos x and justify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

3050

QUESTION ID

S83-M-Q2

TYPE

Short answer

CORRECT ANSWER

0 ≤ x ≤ π: cos decreases from 1 to −1 there, taking each value once

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the standard restricted domain of tan x and justify it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

3051

QUESTION ID

S83-M-Q3

TYPE

Short answer

CORRECT ANSWER

−π/2 < x < π/2: tan increases from −∞ to ∞ there, taking each real value once

GRADING NOTE Form required: domain and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(1/2) and explain why it is not 5π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

3052

QUESTION ID

S83-M-Q4

TYPE

Short answer

CORRECT ANSWER

π/6: arcsin outputs only angles in [−π/2, π/2], and 5π/6 is outside

GRADING NOTE Form required: value and reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arccos(−1/2) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

3053

QUESTION ID

S83-M-Q5

TYPE

Short answer

CORRECT ANSWER

2π/3

GRADING NOTE Form required: exact value in [0, π].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • MAIN • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 1

Restrict a domain to make a function invertible • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d

ANSWER KEY Reduced copy of the question page, with its answer

Give a domain on which $f(x) = x^2$ is invertible.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

3055

QUESTION ID

S83-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 0$ (or $x \leq 0$)

GRADING NOTE Any valid domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q1/5

Why does the restricted function need to be increasing or decreasing?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

3056

QUESTION ID

S83-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Then no output repeats, so each output has one input

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-B1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q2/5

True or false?

A function that is always increasing on an interval is one-to-one there.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

3057

QUESTION ID

S83-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Monotonic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x - 1)^2$ is one-to-one on

A. $-1 \le x \le 3$

B. all reals

C. $x \ge 1$

D. $x \ge 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

3058

QUESTION ID

S83-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x \ge 1$

GRADING NOTE One side of the vertex.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of x^2 on $x \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

3059

QUESTION ID

S83-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$\sqrt{x}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-B1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.d • Set 83 • BACKWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 2

Exact values on the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(π/6) and sin(5π/6).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

3061

QUESTION ID

S83-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/2 and 1/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q1/5

3061

ANSWER KEY Reduced copy of the question page, with its answer

Give cos(2π/3) and cos(4π/3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

3062

QUESTION ID

S83-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1/2 and −1/2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-B2-Q2**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\tan(\pi/4) = \tan(3\pi/4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q3/5

3063

QUESTION ID

S83-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\tan(\pi/4) = 1$ and $\tan(3\pi/4) = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q3/5

3063

ANSWER KEY Reduced copy of the question page, with its answer

Which angle in [0, π] has cos = −1/2?

A. 2π/3

B. π/3

C. 4π/3

D. 5π/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

3064

QUESTION ID

S83-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2π/3

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give $\sin(-\pi/6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

3065

QUESTION ID
S83-B2-Q5

TYPE
Short answer

CORRECT ANSWER
−1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 83 • BACKWARD 2 • Q5/5

SET 83 • HSF-TF.B.6

BACKWARD 3

Periodicity of trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

3067

QUESTION ID

S83-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of tan x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

3068

QUESTION ID

S83-B3-Q2

TYPE

Short answer

CORRECT ANSWER

π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q2/5

True or false?

A periodic function cannot be one-to-one on its whole domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

3069

QUESTION ID

S83-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Values repeat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

sin x = 0 for

A. x = π/2 + kπ

B. no x

C. x = 2kπ only

D. x = kπ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q4/5

3070

QUESTION ID

S83-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. x = kπ

GRADING NOTE Every multiple of π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Name two angles with cos = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

3071

QUESTION ID

S83-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0 and 2π (for example)

GRADING NOTE Any two valid angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-B3-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 83 • BACKWARD 3 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 1

Compose trigonometric functions with their inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(sin(5π/6)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

3073

QUESTION ID

S83-F1-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate sin(arcsin(0.3)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

3074

QUESTION ID

S83-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S83-F1-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q2/5

True or false?

arccos(cos x) = x for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

3075

QUESTION ID

S83-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only for 0 ≤ x ≤ π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arctan(tan(3π/4)) equals

A. −π/4

B. −3π/4

C. 3π/4

D. π/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

3076

QUESTION ID

S83-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −π/4

GRADING NOTE Into (−π/2, π/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain and range of arccos.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

3077

QUESTION ID

S83-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Domain [−1, 1]; range [0, π]

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 83 • FORWARD 1 • Q5/5

SET 83 • HSF-TF.B.6

FORWARD 2

Solve trigonometric equations with inverses • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sin x = 0.4$ on $0 \le x < 2\pi$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

3079

QUESTION ID

S83-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.41 and 2.73

GRADING NOTE Form required: both. arcsin(0.4) and π − arcsin(0.4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\cos x = 0.4$ on $0 \le x < 2\pi$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

3080

QUESTION ID

S83-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.16 and 5.12

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q2/5

True or false?

arcsin gives only one of the solutions of sin x = c on [0, 2π).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

3081

QUESTION ID

S83-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Use symmetry for the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The second solution of $\sin x = c$ in $[0, 2\pi)$ (with $0 < c < 1$) is

A. $\pi + \arcsin c$

B. $-\arcsin c$

C. $\pi - \arcsin c$

D. $2\pi - \arcsin c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

3082

QUESTION ID

S83-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. π − arcsin c

GRADING NOTE Quadrant II.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain how the restricted range of arccos is used to find all solutions of cos x = c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

3083

QUESTION ID

S83-F2-Q5

TYPE

Short answer

CORRECT ANSWER

arccos gives the solution in [0, π]; the other is 2π − arccos c by symmetry.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S83-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 83 • FORWARD 2 • Q5/5

SET 84 • HSF-TF.B.7 • ADVANCED EXTENSION (+)

Solve trigonometric equations in models with inverse functions

MAIN	Algebra II main question	**HSF-TF.B.7**
BACKWARD 1	Evaluate inverse trigonometric functions on restricted domains	**HSF-TF.B.6**
BACKWARD 2	Parameters of sinusoidal models	**HSF-TF.B.5**
BACKWARD 3	Symmetry of unit-circle solutions	**HSF-TF.A.2**
FORWARD 1	Find all real solutions and solve equations that factor	**HSF-TF.B.7**
FORWARD 2	Solve equations with sum and double-angle identities	**HSF-TF.C.9**

SET 84 • HSF-TF.B.7

MAIN

Algebra II main question • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\sin x - 1 = 0$ for $0 \le x < 2\pi$. Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

3086

QUESTION ID

S84-M-Q1

TYPE

Short answer

CORRECT ANSWER

π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3\cos x = -2$ for $0 \le x < 2\pi$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

3087

QUESTION ID

S84-M-Q2

TYPE

Short answer

CORRECT ANSWER

2.30 and 3.98

GRADING NOTE Form required: both. $\arccos(-2/3)$ and $2\pi - \arccos(-2/3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m. Find every time in $0 \le t < 30$ s when $h = 17$ m.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

3088

QUESTION ID

S84-M-Q3

TYPE

Short answer

CORRECT ANSWER

2.5 s and 12.5 s

GRADING NOTE Form required: both. $\sin(\pi t/15) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Harbor depth is $d(t) = 5 + 4\cos(\pi t/6)$ ft. Find every time in $0 \le t < 12$ h when $d = 7$ ft.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

3089

QUESTION ID
S84-M-Q4

TYPE
Short answer

CORRECT ANSWER
2 h and 10 h

GRADING NOTE Form required: both. $\cos(\pi t/6) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\tan x = 3$ for $0 \le x < 2\pi$ to the nearest hundredth.
A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-M-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

3090

QUESTION ID
S84-M-Q5

TYPE
Short answer

CORRECT ANSWER
1.25 and 4.39

GRADING NOTE Form required: both. arctan 3 and arctan 3 + π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-M-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • MAIN • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 1

Evaluate inverse trigonometric functions on restricted domains • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arcsin(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

3092

QUESTION ID

S84-B1-Q1

TYPE

Short answer

CORRECT ANSWER

π/6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-B1-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arccos(1/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

3093

QUESTION ID

S84-B1-Q2

TYPE

Short answer

CORRECT ANSWER

π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

arctan 3 is between 0 and π/2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

3094

QUESTION ID

S84-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Positive input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

arccos(−2/3) is in quadrant

A. II

B. III

C. I

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

3095

QUESTION ID

S84-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. II

GRADING NOTE Range [0, π], negative cosine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate arctan 3 to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

3096

QUESTION ID

S84-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1.25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-B1-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.6 • Set 84 • BACKWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 2

Parameters of sinusoidal models • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

In h(t) = 12 + 10sin(πt/15), what is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

3098

QUESTION ID

S84-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30 s

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite h = 17 as an equation for sin(πt/15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

3099

QUESTION ID

S84-B2-Q2

TYPE

Short answer

CORRECT ANSWER

sin(πt/15) = 1/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$d(t) = 5 + 4\cos(\pi t/6)$ has a maximum of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q3/5

3100

QUESTION ID

S84-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The maximum is 5 + 4 = 9; 5 is the midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Isolating the cosine in 5 + 4cos(πt/6) = 7 gives

A. cos(πt/6) = 1/2

B. cos(πt/6) = 3

C. cos(πt/6) = 2

D. cos(πt/6) = 7/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q4/5

3101

QUESTION ID

S84-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos(πt/6) = 1/2

GRADING NOTE Subtract 5, divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the midline of d(t)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

3102

QUESTION ID

S84-B2-Q5

TYPE

Short answer

CORRECT ANSWER

d = 5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 84 • BACKWARD 2 • Q5/5

SET 84 • HSF-TF.B.7

BACKWARD 3

Symmetry of unit-circle solutions • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with sin = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

3104

QUESTION ID

S84-B3-Q1

TYPE

Short answer

CORRECT ANSWER

π/6 and 5π/6

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with cos = 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

3105

QUESTION ID

S84-B3-Q2

TYPE

Short answer

CORRECT ANSWER

π/3 and 5π/3

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

tan x = c has two solutions in [0, 2π) that differ by π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

3106

QUESTION ID

S84-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Period π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If cos x = c has solution a in [0, π], the other in [0, 2π) is

A. 2π − a

B. π + a

C. −a

D. π − a

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

3107

QUESTION ID

S84-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2π − a

GRADING NOTE Reflection over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give both angles in [0, 2π) with tan = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

3108

QUESTION ID

S84-B3-Q5

TYPE

Short answer

CORRECT ANSWER

π/4 and 5π/4

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 84 • BACKWARD 3 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 1

Find all real solutions and solve equations that factor • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Write all real solutions of $2\sin x - 1 = 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

3110

QUESTION ID

S84-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x = \pi/6 + 2k\pi$ or $x = 5\pi/6 + 2k\pi$ (k an integer)

GRADING NOTE Form required: both families.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2\sin^2 x - \sin x = 0$ for $0 \le x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

3111

QUESTION ID

S84-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0, π, π/6, 5π/6

GRADING NOTE Form required: all four. $\sin x(2\sin x - 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q2/5

True or false?

cos x = 2 has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

3112

QUESTION ID

S84-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q3/5

How many solutions does $\sin(2x) = 1/2$ have in $0 \le x < 2\pi$?

A. 2

B. 4

C. 8

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

3113

QUESTION ID

S84-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

GRADING NOTE 2x runs over two periods.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F1-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\cos^2 x = 1/4$ for $0 \le x < 2\pi$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

3114

QUESTION ID

S84-F1-Q5

TYPE

Short answer

CORRECT ANSWER

π/3, 2π/3, 4π/3, 5π/3

GRADING NOTE Form required: all four.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F1-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 84 • FORWARD 1 • Q5/5

SET 84 • HSF-TF.B.7

FORWARD 2

Solve equations with sum and double-angle identities • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Use sin 2x = 2sin x cos x to solve sin 2x = cos x for $0 \le x < 2\pi$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

3116

QUESTION ID

S84-F2-Q1

TYPE

Short answer

CORRECT ANSWER

π/6, π/2, 5π/6, 3π/2

GRADING NOTE Form required: all four. cos x(2sin x − 1) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $\cos 2x = 1 - 2\sin^2 x$ to solve $\cos 2x = \sin x$ for $0 \le x < 2\pi$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

3117

QUESTION ID

S84-F2-Q2

TYPE

Short answer

CORRECT ANSWER

π/6, 5π/6, 3π/2

GRADING NOTE Form required: all three. $2\sin^2 x + \sin x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(x + \pi/2) = \cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q3/5

3118

QUESTION ID

S84-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sum formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q3/5

cos 2x equals

A. $1 + 2\sin^2 x$

B. $\cos^2 x - \sin^2 x$

C. $\cos^2 x + \sin^2 x$

D. $2\cos x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

3119

QUESTION ID

S84-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $\cos^2 x - \sin^2 x$

GRADING NOTE Double angle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q4/5

Explain why identities are useful for solving equations with different trigonometric functions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S84-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5

3120

QUESTION ID
S84-F2-Q5

TYPE
Short answer

CORRECT ANSWER
They rewrite the equation in one function (or factor it), so inverse functions can be applied.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S84-F2-Q5**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 84 • FORWARD 2 • Q5/5